

Load forecasting for regional integrated energy system based on complementary ensemble empirical mode decomposition and multi-model fusion

Jian Shi, Jiashen Teh^{*}

School of Electrical and Electronic Engineering, Universiti Sains Malaysia (USM), Nibong Tebal 14300, Penang, Malaysia

HIGHLIGHTS

- CEEMD technology reduces the complexity of data samples.
- Multi-model fusion forecasting method improves the forecasting accuracy.
- Sample entropy is used to extract the features of the Intrinsic Mode Function.
- Parameter optimization using particle swarm optimization and genetic algorithm.

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ABSTRACT

The multiple loads of the Regional Integrated Energy System (RIES) possess characteristics of randomness and relatively higher complexity. The current forecasting methods struggle to effectively handle the non-stationary sequence of these multiple loads, leading to less accurate load forecasting. To address this problem, this paper proposes a multi-model fusion prediction method based on Complementary Ensemble Empirical Mode Decomposition (CEEMD), Genetic Algorithm-Long Short Term Memory (GA-LSTM), Radial Basis Fusion-Autoencoder (RBF-AE), and Particle Swarm Optimization-Support Vector Machine (PSO-SVM). First, the load sequence is decomposed into different frequency Intrinsic Mode Functions (IMFs) components using CEEMD. The IMFs components are then grouped based on their zero-crossing rate and Sample Entropy (SE), resulting in three distinct groups: high-, medium-, and low-frequency components. Next, the high-frequency load component, which exhibit strong randomness, are predicted using GA-LSTM. The medium-frequency load component, which have weaker randomness, are predicted using RBF-AE. The smooth and periodic low-frequency load component are predicted using PSO-SVM. The prediction results from these three models are reconstructed to obtain the final predictive value. Finally, experimental results confirm that the forecasting model can effectively handle non-stationary load sequences and demonstrate the highest level of forecasting accuracy.

1. Introduction

The primary objective of the Regional Integrated Energy System (RIES) is to improve the efficiency of various energy sources, including electric, cooling, heating, and others, by introducing greater diversity into the existing energy system. To ensure reliable and cost-effective operation of the RIES, precise load forecasting is essential. However, due to the interdependence among various loads, known as Coupledness, within the RIES, a more sophisticated approach is required to forecast multi-element loads and meet the increased demands [1].

As the energy transition process gains momentum, the study of various types of load forecasting has witnessed a gradual surge. Methods like wavelet neural network, improved particle swarm, and attention-based convolutional neural network have been proposed in literature [2–4] to enhance the accuracy of load forecasting for integrated energy systems. Literature [5] presents a novel short-term load multi-step forecasting method based on long-and short-term time series network. On the other hand, literature [6] proposes a hybrid model called CEEMDAN-SE-TR, which incorporates Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), SE, and Transformer (TR). Literature [7] utilizes the Levy Flight-Particle Swarm

^{*} Corresponding author.

E-mail address: jiashenteh@usm.my (J. Teh).

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Nomenclature*Symbol abbreviation*

BiLSTM	Bidirectional-Long Short Term Memory
BiGRU	Bidirectional Gated Recursive Unit
BPNN	Back Propagation Neural Network
CEEMD	Complementary Ensemble Empirical Mode Decomposition
CEEMDAN	Complete Ensemble Empirical Mode Decomposition with Adaptive Noise
CEEMD-MF	CEEMD and Multi-model Fusion
CNN	Convolutional Neural Network
ELM	Extreme Learning Machine
EMD	Empirical Mode Decomposition
EEMD	Ensemble Empirical Mode Decomposition
GA-LSTM	Genetic Algorithm-Long Short Term Memory
GRU	Gated Recurrent Unit
IchOA	Improved Chimp Optimization Algorithm
IMFs	Intrinsic Mode Functions
ISSA	Improved Sparrow Search Algorithm
IRSA	Improved Reptile Search Algorithm
KELM	Kernel Extreme Learning Machine
LSTM	Long Short Term Memory
LSSVM	Least Squares Support Vector Machine
MAPE	Mean Absolute Percentage Error
PACF	Partial Autocorrelation Function
PSO-SVM	Particle Swarm Optimization-Support Vector Machine
RBF	Radial Basis Function
RBF-AE	Radial Basis Fusion-Autoencoder
RF	Random Forest
RMSE	Root Mean Square Error
RIES	Regional Integrated Energy System
SE	Sample Entropy
SSA	Sparrow Search Algorithm
SVM	Support Vector Machine
TCN	Temporal Convolutional Network
TVFEMD	Time Varying Filtering based Empirical Mode Decomposition
TR	Transformer
VMD	Variational Mode Decomposition
WNN	Wavelet Neural Network

Variables

P_{load}	Total load forecast value
$P_{IMF_{high}}$	The predicted value of high-frequency components
$P_{IMF_{medium}}$	The predicted value of medium-frequency components
$P_{IMF_{low}}$	The predicted value of low-frequency components
$\hat{X}_n$	Normalized load data
X_n	Historical load data
X_{max}	The maximum value in the original power load data
X_{min}	The minimum value in the original power load data
X'_n	The predicted value after denormalization
$\eta(\tau)$	The target signal
$\omega^i(n)$	Gaussian white noise with unit variance
ε_0	The signal to ratio
$IMF_1^z(\tau)$	Each IMF signal obtained by decomposition
$r_1(\tau)$	The τ -th residual.
$IMF(\tau)$	The τ -th IMF component.
E_1	The IMF decomposed by EMD
U	The sum of $IMF(\tau)$
$r_u(\tau)$	The U -th residual
$\varepsilon_u E_u(\omega^u(\tau))$	The U -th signal to ratio which is selected for each stage gaussian white noise
$R(\tau)$	The final residuals
$x(t)$	The original signal

$n_i^+(t), n_i^-(t)$	Two white noise sequences with opposite signs
$p_i^+(t), p_i^-(t)$	Newly obtained signal sequences
$F_{ik}(t), F_{-ik}(t)$	The K th eigenmode components
K	Number of components
$I_k(t)$	The K th total eigenmode component
$R_e(t)$	The final residual component
d	The distance
i, j	$i, j = 1, 2, \dots, N - m + 1$
r	The similar tolerance
$B_i^m(r)$	The ratio of this number to the total number of vectors $N - m$
$num\{\}$	The sum function
m	The dimension
N	The sequence length
$SampEn(m, r, N)$	Sample entropy value
σ_x	The standard deviation of the original data sequence
ε	The variance of the Gaussian function
c_m	The center of the Gaussian function
$\ x_n - c_m\ $	The two-norm calculation of the input sample and variance
w_{ij}	The weight coefficient
h	A low-dimensional hidden variable
$h_{w,b}(x)$	The mapping relationship function
L	The target dimension of RBF-AE
S_1	The cross-entropy loss function
z	The actual value of RBF-AE
$\bar{z}$	The predicted value of RBF-AE
x_i, y_i	Set of training samples
$\varphi(x)$	A nonlinear mapping
W	The weight coefficient of SVM
b	The bias item
ξ_i, ξ_i^*	Slack variables
$K(x_i, x_j)$	The radial basis kernel function
ν	The sensitive loss coefficient
ω	Inertial weight factor
c_1, c_2	Acceleration constants
r_1, r_2	Random numbers
x_{id}	The position of particle
v_{id}	The velocity of particle
p_{id}	The current optimal position of the population searched
p_{gd}	The optimal position particle
u_i	the output value of the sample
$\hat{u}_i$	the predicted value of the SVM model
q	The number of RIES samples
y_λ	The real value of CEEMD-MF
$\hat{y}_\lambda$	The predicted value of CEEMD-MF
λ	Prediction moments of the CEEMD-MF model
Z_j	The zero-crossing rate
j_{zero}	The number of zero-crossing points
J	The length of the signal

Parameters

Max_{iter}	Maximum Iterations
NR	Noise Groups
NE	The number of EEMD Modes
$Nstd$	Noise Standard Deviation
C	The penalty factor of SVM
σ	The kernel parameter
W_f	The weight parameters of the forget gate
b_f	The bias vectors of the forget gate
W_v	Weight parameters of the <i>sigmoid</i> layer
b_v	The bias vectors of the <i>sigmoid</i> layer
w_g	Weight parameters of the <i>tanh</i> layer
b_g	The bias vectors of the <i>tanh</i> layer

W_o	Weight parameters of the output gate
b_o	The bias vectors of the output gate
$\otimes$	The Hadamard product
α_i, α_j and α_i^*, α_j^*	The Lagrange multiplier dual parameters
α	The balance parameter for data fidelity constraint

τ	The time step
R	The number of decomposition layers
DC	Synthetic signal
Ini	The signal initialization
tol	The convergence fault tolerance criterion

Optimization (LF-PSO) technique to optimize a deep learning model for load prediction. Meanwhile, literature [8] proposes a Bidirectional-Long Short Term Memory (BiLSTM) network for heat load prediction, and the Bi-LSTM model can improve the prediction accuracy of heat load in a district heating system by using both past and future weather information. For short-term load prediction, literature [9] introduces a Kernel Extreme Learning Machine (KELM) based method, while literature [10] proposes a relatively simple artificial neural network-based model for short-term load forecasting, which can be used for single-category loads. The aforementioned prediction methods are relatively simple, as they are all based on a single prediction model, which is suitable for handling a single type of load. Furthermore, when confronted with highly fluctuating loads, the predictive capability is inadequate. And these models lack the ability to handle high-dimensional feature vectors. In RIES where electric, cooling and heating loads are interdependent, more complex forecasting models are required to improve accuracy in forecasting.

Predicting raw data directly leads to significant errors. Over the past few years, researchers have introduced signal decomposition techniques to construct advanced deep learning prediction models. The study in [11] proposed a novel hybrid prediction model, which combines Complementary Ensemble Empirical Mode Decomposition (CEEMD), Sample Entropy (SE), Random Forest (RF), Improved Reptile Search Algorithm (IRSA), Extreme Learning Machine (ELM) and BiLSTM. Firstly, the CEEMD technique is applied to break down the initial sequence into a set of more stable components. Secondly, SE segregates these components into high and low-frequency components. The RF algorithm is employed to eliminate redundant features, and BiLSTM and ELM are employed to predict the high and low-frequency components, respectively. The outcomes indicate that the hybrid prediction model exhibits robust predictive capabilities. The Bidirectional Gated Recursive Unit (BiGRU) demonstrates robust capabilities in nonlinear fitting and effective mapping for prediction. The study in [12] proposed a prediction approach referred to as VMD-CNN-BiGRU. In this approach, the first step involves applying Variational Mode Decomposition (VMD) to break down the original sequence. Secondly, Convolutional Neural Networks (CNN) are employed to uncover potential input-output relationship, extracting significant feature correlations. Lastly, prediction is accomplished using Bidirectional Gated Recursive Units (BiGRU). The outcomes indicate that this model exhibits high prediction accuracy.

The load forecasting for RIES proposed in this paper faces challenges in meeting the constantly fluctuating demands of residents and businesses, which makes accurate predictions more difficult. To tackle this issue, literature [13] suggests an integrated load forecasting model based on Bidirectional Generative Adversarial Networks, while literature [14] proposes a short-term multi-energy load forecasting method using a predictive model for multi-task learning. Other studies, such as literature [15–17], propose prediction models based on least squares support vector machine, long short memory neural network model, CNN, and Gated Recurrent Unit (GRU). The aforementioned forecasting model disregards the significant non-stationary characteristics of multivariate loads and lacks comprehensive load processing. Consequently, it directly predicts the loads, resulting in low accuracy. Generally speaking, the model's applicability is limited, and it tends to produce prediction errors when faced with challenging data.

Adequate raw data preprocessing is essential for load forecasting, as it has a direct impact on the model's performance and its ability to provide accurate predictions. Signal decomposition techniques can

assume a pivotal role in predictive data processing, particularly when handling intricate load data featuring complex structures and multi-scale components. The study in [18] proposed a prediction model named Time Varying Filtering based Empirical Mode Decomposition-Partial Autocorrelation Function-Improved Chimp Optimization Algorithm-Bidirectional Gated Recurrent Unit (TVFEMD-PACF-IChOA-BiGRU) is introduced. This model employs TVFEMD to decompose the initial wind speed data. TVFEMD is an adaptive method well-suited for handling nonlinear and non-stationary signals. Experimental results prove that this method inherits the advantages of high frequency separation performance and stability at low sampling rates, and has strong immunity to noise interference and robustness to wind speed data processing. Furthermore, IChOA is applied to refine the weight coefficients, resulting in improved prediction accuracy. The study in [19] proposed a model called Variational Heteroscedastic Gaussian Process Regression (VHGPR). This model is designed with the aim of delivering robust and broadly applicable predictions for wind speed, showcasing high-performance characteristics. The CEEMDAN technique is employed for preprocessing the initial dataset. CEEMDAN effectively addresses mode mixing issues in empirical mode decomposition, exhibits minimal noise interference, and proves suitable for handling intricate nonlinear and non-stationary signals. Empirical findings demonstrate that integrating the CEEMDAN decomposition approach with this prediction model yields enhanced predictive accuracy.

In order to improve the forecasting accuracy, literature [20] employs VMD to decompose the wind speed sequence into various eigenmode components, which are then utilized by the Long Short Term Memory (LSTM) neural network for prediction. However, VMD is only suitable for processing signals with highly concentrated frequencies in a short period of time, which can be attributed to external shocks and natural factors that may affect the power grid. In the case of RIES, the long-term fluctuations of the overall data need to be taken into account, which makes the Empirical Mode Decomposition (EMD) method more appropriate. The EMD process involves screening the original load data by identifying all extreme points of the signal, including the maximum and minimum values, and then taking the average of these points to create a line. This line is then subtracted from the original signal, resulting in a sieving process. The EMD process separates the original signal into different frequency components, and then filters each component to the last remaining monotonic function line. In simpler terms, EMD can be compared to a machine that separates a bunch of mixed coins into several parts based on their value, such as 10 dollars, 1 dollar, 5 cents, and 1 cent. In literature [21], EMD is used. Although each component is consistent with the original sequence after reconstruction, EMD exhibits a modal mixing phenomenon, where the decomposed high-frequency components are non-stationary components. Predicting these components may result in an error. To reduce the impact of non-stationary components generated by decomposition on prediction, Literature [22] employs a set EMD to decompose the component again. However, this approach introduces noise, and the second decomposition generates too many components, making the forecasting process very time-consuming. Therefore, the EMD method suffers from some drawbacks such as modal aliasing and low efficiency, which ultimately leads to low prediction accuracy. Given the aforementioned issues with EMD, this paper proposes the use of the CEEMD method to decompose the original load. Unlike the EMD method that adds a set of white noise to each decomposition, the CEEMD adds a set of positive and negative white noise. This approach can offset the noisy data during decomposition in

both positive and negative directions, which leads to higher efficiency and accuracy of the prediction model.

CEEMD is a method for EMD that adjusts the amplitude locally. It involves breaking down the original signal into various IMFs and a residual term. The front IMFs typically contain high-frequency components, the middle IMFs contain medium-frequency components, and the latter IMFs and RES contain low-frequency components. The same prediction model is used for all the IMFs components obtained via CEEMD decomposition, and the predictions are combined by superimposition to obtain the final prediction value. However, the prediction accuracy achieved by this method is not ideal, particularly when it comes to RIES multi-element load prediction. In the literature [20], LSTM is employed to predict all the IMFs obtained by decomposition. However, LSTM shows poor prediction performance and takes a long time to process low-frequency components [25]. This paper proposes a method to group the IMFs components obtained through CEEMD decomposition and select a suitable sub-prediction model for each group. To improve the effectiveness of grouping, the paper introduces the use of zero-crossing rate and SE as grouping criteria. The zero-crossing rate is a measure of the number of times a signal crosses the zero point in a given time period [23]. In signal processing, it is often used to quantify the number of zero-crossing points in an audio or vibration signal. A zero crossing occurs when the signal changes polarity from positive to negative, or vice versa. By analyzing the zero-crossing rate of a signal, it is possible to distinguish between the high-, medium-, and low-frequency components. Generally, low-frequency components have lower zero-crossing rates because they change more slowly. Medium-frequency components usually have moderate zero-crossing rates, while high-frequency components have high zero-crossing rates because they change rapidly. The interdependent relationship among the three subsystems of electric, cooling and heating in RIES is complex. Therefore, signal analysis may encounter more noise and nonlinearity, making it necessary to consider the SE of each component for effective analysis and distinguishing between high-, medium-, and low frequency components. SE is a technique for assessing the complexity of a time series. It can be used to identify the nonlinear components in the signal sequence based on their complexity. Typically, high-frequency components contain more nonlinear components, while low-frequency components contain fewer nonlinear components [24]. To more accurately differentiate between the high-, medium-, and low-frequency components obtained by CEEMD decomposition, this paper incorporates both the zero-crossing rate and SE.

This paper undertakes a comprehensive examination utilizing the RIES as a framework. It involves breaking down the data on electric, cooling and heating loads within the system, leading to the identification of high-frequency, medium-frequency, and low-frequency components. The high-frequency component embodies the swift variations and immediate fluctuations observed in the electric cooling and heating load data. It mirrors the short-term shifts in demand and showcases the system's ability to respond promptly to these changes. High-frequency components encompass instantaneous power demand fluctuations, as well as rapid changes in cooling and heating loads. They capture transient responses, unexpected events, and both internal and external disturbances affecting the system. These components wield significant influence over the flexibility and stability of system operations. The medium-frequency component pertains to the medium-term alterations and periodic fluctuations observed in the electric cooling and heating load data. It signifies the seasonal, periodic, and repetitive changes in the load. Medium-frequency components encompass seasonal variations in electricity, cooling, and heating loads, as well as cyclical patterns and distinctions between working and non-working days. They bear significant implications for system planning, scheduling, and resource allocation. The low-frequency component captures the long-term trends and gradual patterns in the electric cooling and heating load data. It reflects the impacts of long-term economic development, population growth, and structural changes in the load, among other factors. Low-frequency components encompass long-term trends, annual variations, economic

cycles, and more in electricity, cooling, and heating loads. They play a crucial role in system planning, long-term energy demand forecasting, investment decision-making, and policy development.

This paper chooses the suitable forecasting model to predict the IMFs of three distinct frequencies. In accordance with literature [25], a load forecasting method is proposed that combines the multiple linear regression forecasting method and LSTM. This method dissects the load sequence into different frequency components and employs LSTM to predict high-frequency components that exhibit strong randomness. The outcomes indicate a higher level of prediction accuracy. LSTM is a neural network that is well suited for time series data due to its long-term memory unit, which can capture long-term dependencies in sequential data. It is appropriate for handling high-frequency data with time series characteristics. High-frequency data often exhibit temporal correlations, and LSTM is a suitable model for processing and predicting such data. Given the strong time series characteristics present in the RIES multivariate load data, the decision has been made in this paper to utilize LSTM for the prediction of high-frequency components.

A prediction method incorporating variational mode decomposition-wavelet transform and principal component analysis-backpropagation screening-radial basis function was proposed in the literature [26]. By decomposing and denoising the raw data sequence, the method aims to extract a medium-frequency component that exhibits relative stability. This extracted component is subsequently fed into a Radial Basis Function (RBF) neural network for prediction purposes. Notably, the final prediction results obtained through this approach demonstrate a high level of accuracy. The RBF neural network is a nonlinear model that employs radial basis functions to determine the weighting of its output. It is particularly effective in high-dimensional spaces. In the context of RIES multivariate load forecasting, the interdependence between electric, cooling, heating loads and meteorological factors plays a vital role in predicting multivariate loads. Understanding and incorporating the coupling characteristics among these factors are crucial for accurate multivariate load forecasting in the RIES framework. In order to make accurate predictions, predictive models necessitate the use of high-dimensional feature vectors. Consequently, RBF neural network is the best choice to predicting medium-frequency components.

The time series forecasting model described in literature [27] combines the Wavelet Neural Network (WNN) and Support Vector Machine (SVM) techniques to improve predictions. The process begins by applying wavelet decomposition to a time series, separating it into high- and low-frequency components. The high-frequency components are fitted using a WNN model, while a SVM model is employed to predict the low-frequency components. The sub-model's predicted value is subsequently reconstructed, resulting in a highly accurate final prediction. The literature indicates that SVM demonstrates remarkable generalization ability and robustness when it comes to predicting low-frequency components. The IMF obtained through RIES multivariate load decomposition often contains numerous outliers and noise, and SVM is adept at mitigating the impact of these factors on prediction outcomes. Consequently, this paper employs SVM for the prediction of low-frequency components.

This paper proposes CEEMD and Multi-model Fusion (CEEMD-MF) prediction methods. The main contributions of this paper are:

- (i) The distinctive features exhibited by RIES multiple loads, particularly their significant fluctuations, are explored in this paper by drawing from the background of RIES. A novel data processing technique called frequency grouping is introduced, which is specifically tailored to address these characteristics. The proposed method focuses on a sequence of non-stationary IMFs and effectively partitions them into three distinct frequency components by combining SE and zero-crossing rate. Multi-scale features in load data are enhanced, noise is eliminated, and the precision and dependability of predictions are enhanced through the use of this method.

(ii) A multi-model fusion forecasting method is proposed. The method takes into consideration the coupling characteristics of electric, cooling and heating loads and utilizes three different models to predict IMF at various frequencies. The prediction model is then subjected to further optimization to improve its performance. Subsequently, the prediction values generated by these sub-models are combined and reconstructed to derive the final prediction value.

The paper is structured as follows: The framework of CEEMD-MF is outlined in Section II, and its forecasting method is introduced. Furthermore, GA-LSTM, PSO-SVM, RBF-AE, CEEMD, and SE are presented. In Section III, the data is processed and decomposed, and the resulting IMF components are categorized as high-, medium-, and low-frequency. The simulation experiment is then conducted from four

different perspectives. Firstly, the impact of CEEMD on accuracy is evaluated by comparing it with LSTM, GA-LSTM, RBF-AE, and PSO-SVM. Secondly, the advantages of CEEMD are analyzed and compared with EEMD-MF and VMD-MF. The third point involves analyzing the impact of multi-model fusion on accuracy. This is done by comparing it with CEEMD-LSSVM, CEEMD-GA-LSTM, and CEEMD-PSO-SVM. Finally, to assess the reliability of the model, a comparison is made with ELM, BPNN, GRU, TCN, ISSA-LSSVM, and CEEMD-SSA-LSSVM. The conclusion is provided in Section IV.

2. Methodology

The proposed prediction model in this paper utilizes CEEMD technology, Zero-crossing rates, SE, GA-LSTM, PSO-SVM, and RBF-AE to create a multi-model fusion approach for forecasting. The process

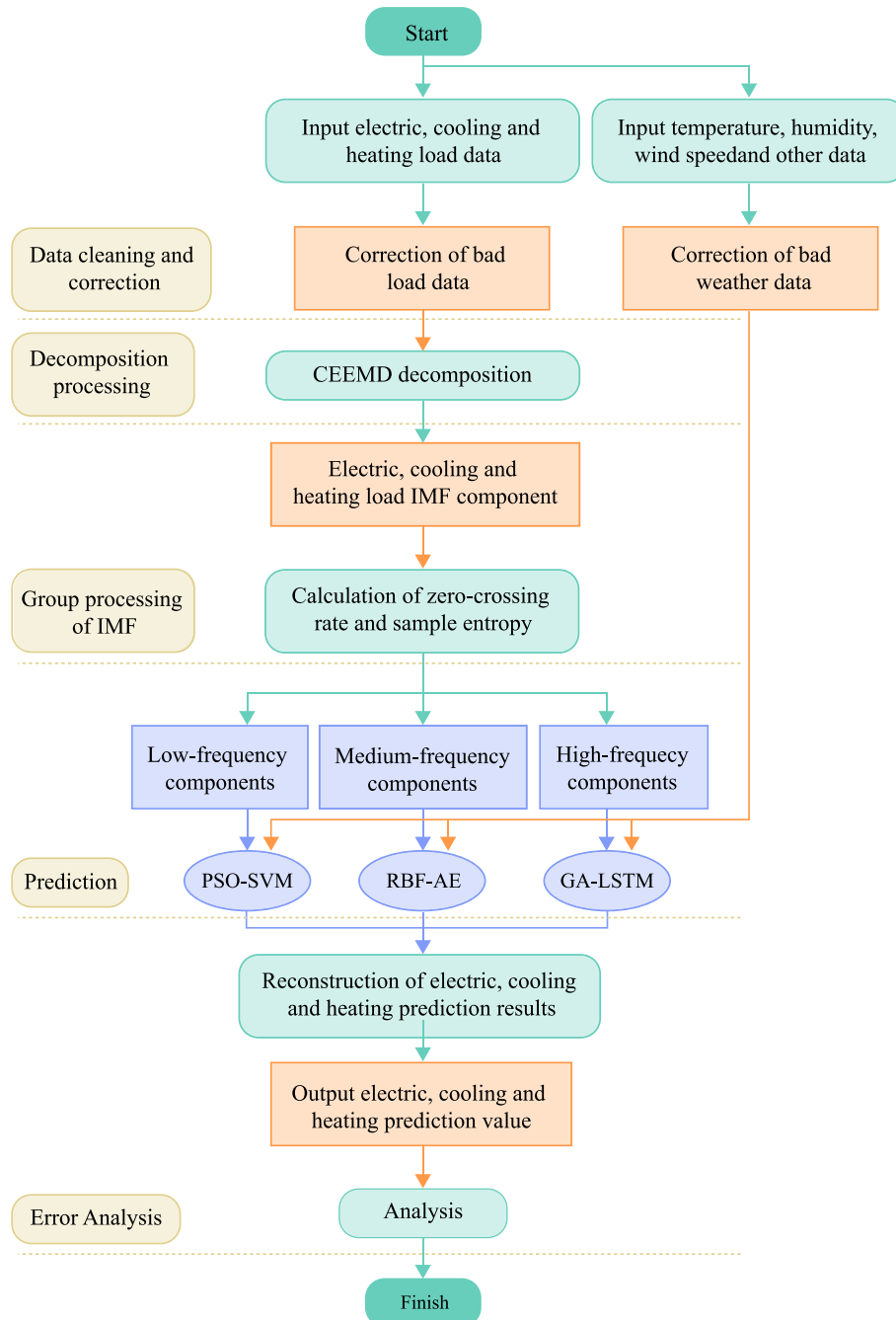


Fig. 1. The flow chart of the CEEMD-MF prediction model.

primarily involves the following steps:

Step 1: The model takes in electric, cooling and heating load data and meteorological data, and performs data validation and error correction to ensure the accuracy and reliability of the input data.

Step 2: The procedure entails applying CEEMD to decompose the electric, cooling and heating load data, calculate the zero-crossing rate and SE of the obtained IMF, and grouping them to obtain high-, medium-, and low-frequency components.

Step 3: For these three different frequency components, PSO-SVM, RBF-AE and GA-LSTM models are introduced to predict them respectively, and finally the prediction results of the three models are reconstructed to obtain the final prediction value. Following this, an error analysis is conducted to assess the precision of the prediction.

The CEEMD decomposition guarantees that all the IMFs are aligned on the same time axis, and by combining them, the complete load curve can be reconstructed. The prediction results obtained from the three prediction models can be directly overlaid on the final prediction result using the following equation:

$$P_{load} = P_{IMF_{high}} + P_{IMF_{medium}} + P_{IMF_{low}} \quad (1)$$

In (1): P_{load} represents the total load forecast value. $P_{IMF_{high}}$ represents the predicted value of high-frequency components. $P_{IMF_{medium}}$ represents the predicted value of medium-frequency components. $P_{IMF_{low}}$ represents the predicted value of low-frequency components.

In RIES multi-element load forecasting, the high-, medium-, and low-frequency are considered equally important. Therefore, the final forecast value can be obtained through a direct linear superposition of these frequencies. Furthermore, as the dataset is divided into different frequency components, distributed computing techniques can be employed to alleviate the computational burden. Fig. 1 illustrates the flow chart of the prediction model.

2.1. Data processing

Data preprocessing is necessary for load forecasting. The following are the processing steps:

- (i) Data cleaning: In the initial step of data cleaning, the original dataset is thoroughly processed to eliminate any potential errors, anomalies, or missing data. This process may involve handling missing values, addressing duplicate entries, and correcting erroneous data, among other tasks. By doing so, the data is refined and prepared for further analysis or modeling. Specifically, this phase involves three major steps. 1) The process of handling missing values involves utilizing the imputation method to replace any missing values, specifically by selecting the mean value for imputation. 2) Outlier handling, Outlier handling involves dealing with data points that exhibit notable deviations from the rest of the observations. In this paper, statistical techniques are applied to identify outliers, and subsequent corrections are made to address them. 3) Standardizing Units: The electric, cooling and heating data sets contain inconsistent units, which are then harmonized to a uniform unit of KW/h. This standardization guarantees data consistency and facilitates effective data comparisons.
- (ii) Data normalization: In order to ensure consistent predictions and avoid the disproportionate influence of various features, it is necessary to normalize the input data prior to making predictions.

Normalize processing load data. Use the maximum value method to normalize the load data, and the processing formula is (2):

$$\hat{X}_n = \frac{X_n - X_{min}}{X_{max} - X_{min}} \quad (2)$$

In (2): $\hat{X}_n$ is the normalized load data, X_n is the historical load data. X_{max} is the maximum value in the original load data, X_{min} is the minimum value in the original load data.

Formula (3) can be used to denormalize the output of the prediction model and obtain the predicted value.

$$X'_n = \hat{X}_n(X_{max} - X_{min}) + X_{min} \quad (3)$$

In (3): X'_n is the predicted value after denormalization.

2.2. Load forecasting evaluation index

The evaluation criteria selected in this paper are the Mean Absolute Percentage Error (MAPE) and the Root Mean Square Error (RMSE). MAPE represents the average relative error between the predicted and actual values, while RMSE is the square root of the ratio of the squared deviation between predicted and actual values to the predicted range. Smaller values of both indicators indicate better prediction performance. The formulas for these two indicators are given below:

$$RMSE = \sqrt{\frac{1}{q} \sum_{\lambda=1}^q (y_{\lambda} - \hat{y}_{\lambda})^2} \quad (4)$$

$$MAPE = \frac{1}{q} \sum_{\lambda=1}^q \left| \frac{y_{\lambda} - \hat{y}_{\lambda}}{y_{\lambda}} \right| \times 100\% \quad (5)$$

In (4) and (5): q is the number of samples involved in the calculation. y_{λ} is the real value at time λ . $\hat{y}_{\lambda}$ is the predicted value at time λ .

2.3. Complementary ensemble empirical mode decomposition

CEEMD is a signal analysis technique that utilizes white noise to aid in the decomposition process. While this approach may cause some disruptions in the smooth signal pulses, it ultimately enhances the computational efficiency of the overall decomposition. Then $\eta(\tau)$ is set as target signal, and the signal added to white noise is $\eta(\tau) + \varepsilon_0 \omega^i(n)$. $\omega^i(n)$ represents zero mean Gaussian white noise with unit variance. ε_0 represents the signal to ratio which is selected for each stage. The specific steps are as follows [28]:

Step1: By subjecting $\eta(\tau) + \varepsilon_0 \omega^i(n)$ to B rounds of EMD decomposition, we obtain IMF_1 as the resulting signal:

$$IMF_1 = \frac{1}{B} \sum_{z=1}^B IMF_1^z(\tau) \quad (6)$$

In (6): $IMF_1^z(\tau)$ represents for each IMF signal obtained by decomposition.

Step2: Calculate the first-order residual r_1 :

$$r_1(\tau) = \eta(\tau) - IMF_1(\tau) \quad (7)$$

In (7): $r_1(\tau)$ represents the τ -th residual. $IMF_1(\tau)$ represents the τ -th IMF component.

Step3: Decompose $r_1(\tau) + \varepsilon_1 E_1(\omega^i(\tau))$, $i = 1, 2, 3, \dots, b$. In CEEMD, the IMF_2 signal is derived from the first intrinsic modal function:

$$IMF_2(\tau) = \frac{1}{B} \sum_{i=1}^b E_1(r_1(\tau) + \varepsilon_1 E_1(\omega^i(\tau))) \quad (8)$$

In (8): E_1 represents the IMF decomposed by EMD. $\varepsilon_1 E_1(\omega^i(\tau))$ represents the signal to ratio which is selected for each stage gaussian white noise.

Step4: The u -th residual r_u is obtained:

$$r_u(\tau) = r_{u-1}(\tau) - IMF_u(\tau) \quad (9)$$

In (9): u denotes the sum of $IMF(\tau)$. $r_u(\tau)$ represents the u -th residual. $r_{u-1}(\tau)$ represents the $u-1$ -th residual.

Step5: Decompose $r_u(\tau) + \varepsilon_u E_u(\omega^u(\tau))$. In CEEMD, the $IMF_{(u+1)}$ signal

is obtained from the first intrinsic mode function:

$$IMF_{u+1}(\tau) = \frac{1}{B} \sum_{i=1}^B E_1(r_u(\tau) + \varepsilon_u E_u(\omega^u(\tau))) \quad (10)$$

In (10): $\varepsilon_u E_u(\omega^u(\tau))$ represents the u -th signal to ratio which is selected for each stage gaussian white noise.

Step6: Repeat steps (7) and (8) until further decomposition of the residual signal is no longer possible.

$$R(\tau) = \eta(\tau) - \sum_{u=1}^u IMF_u(\tau) \quad (11)$$

In (11): $R(\tau)$ represents the final residuals.

Step7: Thus, the target signal $\eta(\tau)$ can be represented as:

$$\eta(\tau) = \sum_{u=1}^u IMF_u(\tau) + R_\tau \quad (12)$$

By utilizing the above formula, we can obtain the complete CEEMD decomposition signal, and accurately reconstruct the original signal.

2.4. Sample entropy

Sample entropy is an improvement upon approximate entropy [29]. Given a data sequence $x(n) = \{x(1), x(2), \dots, x(N)\}$ of a certain length, the steps for calculating sample entropy are as follows:

Step1: Composing the sequence $x(n)$ into a vector of dimension m in order, then:

$$X(i) = [x(i), x(i+1), \dots, x(i+m-1)] \quad (13)$$

In (13): $i = 1, 2, \dots, N - m + 1$.

Step2: Define the distance $d[X(i), X(j)]$ as the value with the largest difference between the corresponding elements of $X(i)$ and $X(j)$, $j = 1, 2, \dots, N - m + 1$, that is:

$$d[X(i), X(j)] = \max_{0 \leq k \leq m-1} |x(i+k) - x(j+k)| \quad (14)$$

In (14): The difference between $X(i)$ and other corresponding elements in $X(j)$ is less than d , and for each i , find the distance $d[X(i), X(j)]$ between $X(i)$ and $X(j)$.

Step3: Count the number of $d[X(i), X(j)] < r$, r is the similar tolerance, $r > 0$. The ratio of this number to the total number of vectors $N - m$ is $B_i^m(r)$, then we have:

$$B_i^m(r) = \frac{1}{N - m} \text{num}\{d[X(i), X(j)] < r\} \quad (15)$$

In (15): $\text{num}\{\}$ is the sum function.

Step4: Take the average of all i to get $B^m(r)$:

$$B^m(r) = \frac{1}{N - m + 1} \sum_{i=1}^{N-m+1} B_i^m(r) \quad (16)$$

Step5: Increase the dimension to $m + 1$, repeat steps 1–4 to get $B_i^{m+1}(r)$ and $B^{m+1}(r)$.

Step6: The theoretical value of sample entropy for a random sequence can be calculated as:

$$\text{SampEn}(m, r, N) = -\ln[B^{m+1}(r)/B^m(r)] \quad (17)$$

In (17): m represents the dimension. r represents the similarity tolerance. N represents the sequence length.

Research has shown that the value of $\text{SampEn}(m, r, N)$ depends on the embedding dimension m and the similarity tolerance r . The research findings indicated that selecting values of m equal to 1 or 2, and setting r within the range of 0.1 to 0.25 times σ_x (the standard deviation of the original data sequence), resulted in enhanced statistical characteristics of the sample entropy measurement. In this paper, the values of $m = 2$ and $r = 0.2\sigma_x$ were selected.

2.5. Genetic algorithm optimizing long short term memory network

2.5.1. LSTM neural network

The LSTM network is an advanced version of the Recursive Neural Network (RNN) model [30]. Instead of using conventional neurons in the hidden layer, it utilizes memory cells as specialized units to store relevant information while discarding irrelevant data. In addition, it incorporates three gating structures: the input gate, output gate, and forget gate, which overcome the issues of gradient vanishing and exploding in traditional RNN training. Consequently, it is a highly appropriate option for time-series prediction tasks. The internal structure of the LSTM network is depicted in Fig. 2. The specific meanings represented by each variable in Fig. 2 are shown in Table 1.

In the LSTM model, each memory cell features an input gate, an output gate, and a forget gate. The input gate regulates the likelihood of fresh data being entered in the current phase, thereby controlling the intensity of new input. The input of new information is controlled jointly by the *sigmoid* layer and the *tanh* layer. On the other hand, the output gate manages the amount of information that is visible from the present state to the next phase. It conceals the hidden state information, thus generating a new hidden layer. The forget gate involves multiplying the hidden output of the previous stage and the input of the current stage by the weight and activating the state of the memory cells in the previous stage. This action results in determining the amount of memory that the memory cells should retain in the previous stage, thus selecting the historical information. If the forget gate is entirely closed, that is, the output of the forget gate is $f_t = 0$, the historical data would have no impact on the current state. The calculation process of the LSTM basic unit is as follows:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (18)$$

$$v_t = \sigma(W_v[h_{t-1}, x_t] + b_v) \quad (19)$$

$$g_t = \tanh(w_g[h_{t-1}, x_t] + b_g) \quad (20)$$

$$O_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (21)$$

$$C_t = \sigma(C_{t-1} \otimes f_t + v_t \otimes g_t) \quad (22)$$

$$h_t = O_t \otimes \tanh(C_t) \quad (23)$$

In (18): t represents the moment, W_f represents the weight parameters of the forget gate, b_f represents the bias vectors of the forget gate. In (19): W_v represents weight parameters of the *sigmoid* layer, b_v represents the bias vectors of the *sigmoid* layer. In (20): w_g represents weight parameters of the *tanh* layer, b_g represents the bias vectors of the *tanh* layer. In (21): W_o represents weight parameters of the output gate, b_o represents the bias vectors of the output gate. In (22) and (23): $\otimes$ represents the *Hadamard* product, which is the product of the

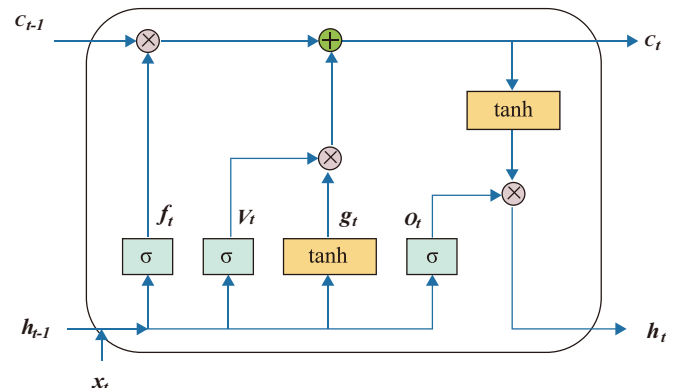


Fig. 2. Structure of LSTM unit.

Table 1
LSTM variables signifying.

Variables	Signifying
h_t	The hidden state at time t
h	The number of hidden units
$\sigma, \tanh$	Activation functions
x_t	The input at time t
h_{t-1}	The output at time $t-1$
f_t	Forget gate
v_t	Input gate
O_t	Output gate
g_t	Candidate value of cell state
C_t	Memory cell at time t
C_{t-1}	Memory cell at time $t-1$

corresponding elements of the two vectors.

Fig. 3 shows the network structure using two layers of LSTM and one fully connected layer, where l is the sequence length.

This paper utilizes the LSTM model to forecast the high-frequency component, and evaluates its advantages and disadvantages in relation to the physical interpretation of the high-frequency component.

The following analysis highlights the strengths and limitations of the LSTM model:

Strengths:

- Long-term dependency modeling: LSTM models excel at effectively capturing and modeling long-term dependencies in time series data by utilizing memory cells and gating mechanisms. This capability proves valuable in predicting rapid changes and transient fluctuations within high-frequency components.
- Nonlinear modeling capabilities: LSTM models possess strong nonlinear modeling capabilities, enabling them to learn intricate patterns and relationships within time series data. This attribute enables better adaptation to nonlinear features present in high-frequency components and facilitates the capture of complex patterns and regularities within the data.
- Processing sequence data: The LSTM model is especially well-suited for handling sequential data and proficiently manages the timing of input data. This capability allows it to model and predict consecutive time points within high-frequency components, making it suitable for time series forecasting tasks involving load data.

Limitations:

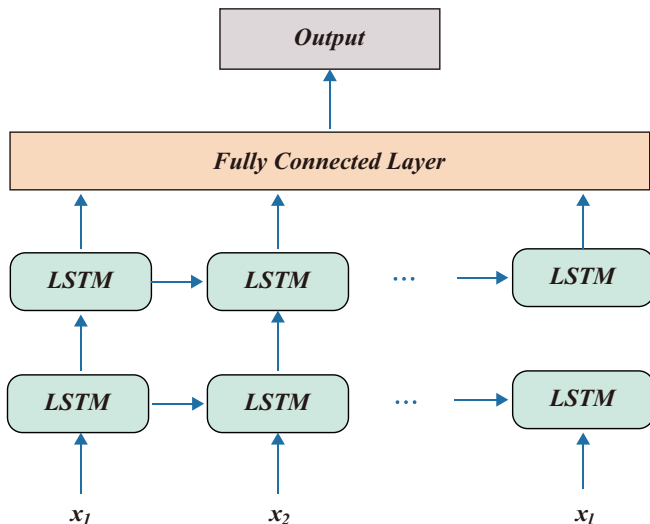


Fig. 3. Structure of LSTM network.

- Parameter adjustment and overfitting: The LSTM model entails several adjustable parameters, such as the number of memory units and learning rate. Improper parameter settings can result in diminished model performance or overfitting issues.
- Lengthy training time: LSTM models typically require a significant amount of time for training, particularly when dealing with large-scale datasets and complex model architectures. This may contribute to increased time expenditures during model development and optimization.

In summary, the LSTM model offers several advantages for predicting high-frequency components and excels at capturing long-term dependencies and nonlinear patterns in time series data. Nevertheless, it is crucial to carefully fine-tune the model parameters to prevent overfitting and ensure an adequate number of data samples for training and prediction.

To demonstrate the suitability of LSTM for high-frequency scenarios, a relevant instance is presented wherein a randomly chosen high-frequency curve is used to predict the last 100 data points. In this study, the error analysis is conducted using RMSE and MAPE metrics, as depicted in Fig. 4. It achieves the lowest MAPE and RMSE values of 2.45% and 21.75, respectively. According to the findings presented in Appendix Fig. A1, LSTM exhibits the highest degree of fitting when assessing the high-frequency components among the models. LSTM's architecture comprises a memory unit and a gating mechanism, empowering it to effectively address temporal dependencies in high-frequency components and capture swift changes in such components. In comparison to other predictive models, LSTM possesses distinct advantages in predicting high-frequency components, consistently delivering superior results. As a result, this paper adopts LSTM as the chosen method for processing high-frequency components.

2.5.2. GA-LSTM model

The genetic algorithm is an optimization method that mimics the biological evolution mechanism in nature by conducting a random global search. It is a parallel and efficient search method that can adaptively control the search process while acquiring and accumulating knowledge about the search space during the search process. The genetic algorithm takes the population as the solution to the problem and generates a new population through selection, crossover, and mutation operations. It relies on a fitness function as its decision criterion and gradually moves closer to the population with the best fitness value during the search cycle, eventually obtaining the optimal solution to the problem.

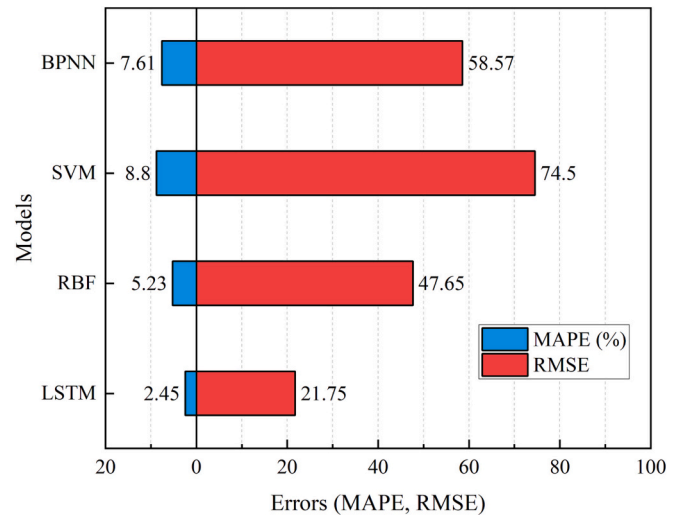


Fig. 4. Error analysis of four models at high-frequency.

Table 2 displays the parameters that require optimization in LSTM. The following are the steps for GA-LSTM prediction:

Step1: To obtain and prepare the necessary data, this involves acquiring and preprocessing the data, as well as conducting screening to filter out any irrelevant data and supplementing missing values.

Step2: The first step is to provide input data and then establish the parameter settings for the genetic algorithm, including the DNA and chromosome lengths, population size, crossover and mutation rates, as well as the number of iterations.

Step3: V populations are generated randomly, and then the genetic algorithm proceeds to calculate the fitness value of each individual after applying crossover and mutation operators. The algorithm then selects the optimal solution based on the fitness value for each generation.

Step4: The genetic algorithm then generates a new population and repeats the process until the number of iterations reaches the maximum value. The resulting optimal solution is then used as a hyperparameter for the LSTM model. The GA-LSTM model is trained using this hyperparameter.

Step5: Make predictions and output predicted values.

2.6. Radial basis function and autoencoder fusion prediction model

2.6.1. RBF neural network

The RBF neural network, as depicted in Fig. 5, is a conventional feed-forward neural network that excels at fitting nonlinear continuous functions. It consists of input layer ($X_1 \sim X_N$), hidden layer ($C_1 \sim C_m$), and output layer Y . The network's distinctive feature is that the activation function of the hidden layer nodes is positioned as the radial basis function, which offers several advantages, including fast convergence speed, high fitting accuracy, and a simple network structure [31].

In the RBF structure depicted in Fig. 5, the activation function in the hidden layer typically employs a radial basis function.

$$R(x_n - c_m) = \exp\left(-\frac{1}{2\varepsilon^2}\|x_n - c_m\|^2\right) \quad (24)$$

In (24): x_n is the n th input parameter, c_m is the center of the Gaussian function, ε is the variance of the Gaussian function, $\|x_n - c_m\|$ is the two-norm calculation of the input sample and variance, and finally the RBF neural network result y :

$$y = \sum_{i=1}^m w_{ij} \exp\left(-\frac{1}{2\varepsilon^2}\|x_n - c_m\|^2\right) \quad (25)$$

In (25): w_{ij} is the weight coefficient, $i = 1, 2, 3, \dots, n$.

This paper adopts the RBF model for forecasting the medium-frequency component, and assesses its strengths and weaknesses in conjunction with the physical interpretation of the medium-frequency component. The ensuing analysis delves into the strengths and limitations of the RBF model:

Strengths:

- (i) Adaptation to periodic changes and seasonality: The intermediate frequency component aligns with medium-term changes and seasonal fluctuations in the electric, cooling and heating load data, and the RBF model demonstrates better adaptability to these periodic variations. By employing suitable basis functions and parameter configurations, the RBF model can effectively

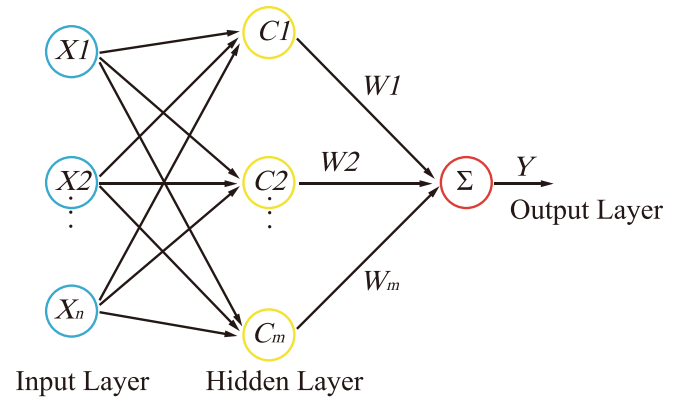


Fig. 5. RBF neural network structure.

capture the seasonal pattern within the medium-frequency component and enhance prediction accuracy.

- (ii) Nonlinear modeling capability: The RBF model is endowed with robust nonlinear modeling abilities, allowing it to learn and depict the nonlinear modes and relationships within intricate intermediate frequency components. This empowers the model to adeptly handle nonlinear features and complex relationships within the medium frequency components, ultimately enhancing prediction accuracy.
- (iii) Flexibility and generalization capacity: The RBF model exhibits remarkable flexibility, enabling it to adaptively learn the characteristics and patterns of data. It possesses robust generalization capabilities, allowing it to make accurate predictions for unfamiliar data and adapt to diverse scenarios and evolving load demands within the RIES.

Limitations:

- (i) Training data requirements: The RBF model necessitates a sufficient amount of training data to acquire precise model parameters. Predicting the medium-frequency component may require incorporating data from multiple seasonal periods, thereby underscoring the importance of gathering ample historical data to support the model training process.
- (ii) Computational complexity: RBF models exhibit high computational complexity during both training and prediction stages, particularly when dealing with large-scale datasets and intricate model structures. Consequently, this may result in increased computational and time costs associated with the model.

Overall, the RBF model offers distinct advantages for forecasting the medium-frequency component of the electric, cooling and heating load within the RIES. It effectively adapts to periodic changes and nonlinear characteristics, showcasing flexibility and generalization capabilities. However, careful consideration must be given to basis function selection and parameter adjustment, while ensuring an ample supply of training data to support model training. In practical applications, refining the model settings through experience and practice can yield more accurate and reliable predictions for medium frequency components.

To demonstrate the suitability of RBF for handling medium frequency components, a random medium frequency curve is chosen for predicting the last 100 points. The evaluation of prediction accuracy is conducted using RMSE and MAPE as error metrics for analysis. Based on the results depicted in Appendix Fig. A2, it is clear that RBF outperforms the other models in terms of fitting degree for the medium frequency components. Based on the error analysis depicted in Fig. 6, RBF achieves the lowest MAPE and RMSE values of 2.12% and 4.58, respectively. While the fluctuation of intermediate frequency components may not be

Table 2

LSTM needs optimized hyperparameters.

Optimization parameters	Range of initial values
LSTM algorithm layers	[1, 3]
The number of neurons in the LSTM layer	[32, 256]
The number of fully connected layers	[1, 3]
The number of neurons in each fully connected layer	[32, 256]

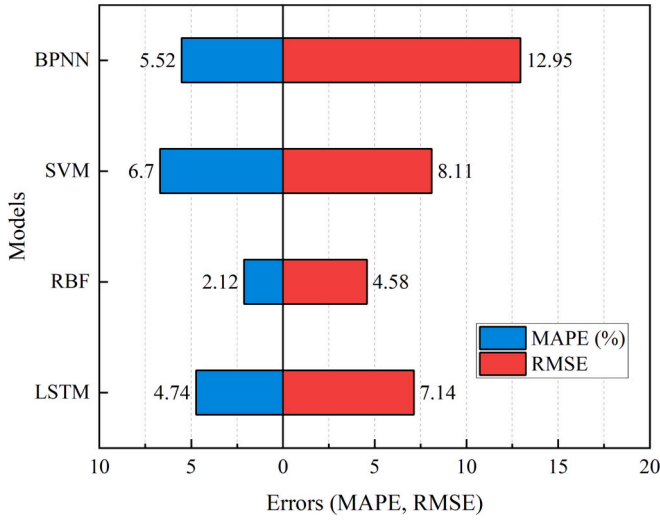


Fig. 6. Error analysis of four models at medium-frequency.

as pronounced as that of high-frequency components, it still encompasses a significant amount of nonlinear elements. Effectively handling these nonlinear components within the medium frequency component is crucial for enhancing prediction accuracy. The RBF network employs the radial basis function as its activation function, commonly utilizing the Gaussian function. The Gaussian function possesses nonlinear characteristics and is capable of performing nonlinear mapping on input data. By leveraging this nonlinear activation function, the RBF network can effectively capture the nonlinear relationship within the input data. Moreover, the RBF network's local approximation capability enables it to handle intermediate frequency components more effectively. Hence, in this paper, RBF is employed as the chosen method for predicting medium frequency components.

2.6.2. RBF-AE model

The self-encoder AE is designed to tackle complex high-dimensional computations, comprising encoders and decoders. As an unsupervised learning technique, it utilizes input data as supervision and back-propagation method to establish a relationship between the input and target values. This helps guide the neural network to learn a mapping relationship. In the self-encoder network, the encoder can be represented as $h = f(x)$, which encodes the input x into a low-dimensional hidden variable h . This aids the neural network in efficiently learning the most valuable information features [32,33]. On the other hand, the decoder is expressed as $h = g(h)$, which restores the hidden variable h in passed neural network to the original dimension. As a result, the output r can be restored to be similar to the input x dimension.

The objective of using the network is to learn the mapping relationship function $h_{w,b}(x) \approx \tilde{x}$, essentially fitting an identity function through network training, so that the output is nearly identical to the input x . To construct an autoencoder, the following processes are typically required: 1). Designing the encoder, 2). Designing the decoder, and 3). Building a loss function.

Since autoencoders are unsupervised learning models, they do not require labeled data for training. The autoencoder can be trained to replicate the input as output, and then the structure of the autoencoder and the input data of the RBF are utilized for output, resulting in $Y = X$. The RBF-AE network's structure, as a fusion autoencoder, is depicted in Fig. 7.

In Fig. 7, the RBF-AE network structure can achieve dimension compression by mapping high-dimensional data into a low-dimensional representation if $n > L$, where L is the target dimension. If $n = L$, RBF-AE achieves equal-dimensional feature expression, and if $n < L$, it realizes sparse expression, which means high-dimensional feature expression of

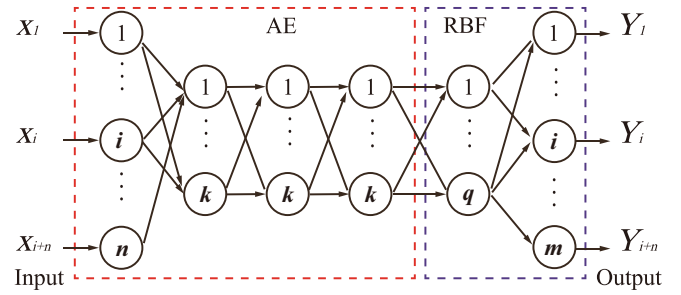


Fig. 7. RBF-AE network structure diagram.

the original data. The cross-entropy loss function is used in the RBF-AE algorithm model. Compared to the quadratic loss function, the cross-entropy loss function can avoid the need to derive the activation function in the gradient update. This eliminates the limitation of the activation function and allows for larger gradients with larger real errors, which speeds up convergence and shortens training time. The cross-entropy function can be expressed as:

$$S_1 = -z \log \bar{z} + (z - 1) \log(1 - \bar{z}) \quad (26)$$

In (26): S_1 represents the cross-entropy loss function, where z is the actual value and $\bar{z}$ is the predicted value. The RBF-AE model prediction process consists of the following steps:

Step 1: Data acquisition and preprocessing, including dividing the dataset into training and testing sets.

Step 2: Input data and normalize the training set. Detect data features, reduce dimensionality, determine the number of RBF hidden layer nodes, and train the model.

Step 3: Use the trained algorithm model to predict and output the predicted value.

2.7. Particle swarm optimization for support vector machine

2.7.1. Support vector machine model

For a given set of training samples (x_i, y_i) , where $x_i \in \mathbb{R}^n$ is the input and $y_i \in \mathbb{R}^n$ is the output, the support vector machine regression theory achieves nonlinear transformation of the sample space by introducing a nonlinear mapping $\varphi(x)$ [34]. For nonlinear problems, the estimated function $f(x)$ can be transformed into the following form:

$$f(x) = W \cdot \varphi(x) + b \quad (27)$$

In (27): W is the weight coefficient. b is the bias item. ξ_i, ξ_i^* are slack variables. And ξ_i, ξ_i^* are both > 0 , then the problem is transformed into a dual problem, and the optimization problem is solved:

$$\min_{W, b} \frac{1}{2} \|W\|^2 + C \sum_{i=1}^l (\xi_i + \xi_i^*) \quad (28)$$

In (28): C is the penalty factor. α_i, α_j and α_i^*, α_j^* are the Lagrange multiplier dual parameters, this quadratic programming optimization problem with inequality constraints can be transformed into the following problem:

$$\max \left[-\frac{1}{2} \sum_{i,j=1}^n (\alpha_i^* - \alpha_i) (\alpha_j^* - \alpha_j) K(x_i, x_j) + \sum_{i=1}^l \alpha_i^* (y_i - v) - \sum_{i=1}^n \alpha_i (y_i - v) \right] \quad (29)$$

In (29): $K(x_i, x_j)$ is the radial basis kernel function. v is the sensitive loss coefficient. The radial basis function is chosen as the kernel function for SVM, and it is defined as:

$$K(x_i, x_j) = \exp \left(-\frac{\|x_i - x_j\|^2}{2\sigma^2} \right) \quad (30)$$

In (30): the kernel parameter σ needs to be determined, and by solving the optimization problem, the regression function of the support vector machine can be obtained as follows:

$$f(x) = \sum_{i,j=1}^n (\alpha_i - \alpha_i^*) K(x_i, x_j) + b \quad (31)$$

The selection of the kernel parameter σ greatly affects the prediction accuracy of the SVM model. Thus, in order to optimize the SVM model, the kernel parameter σ and penalty factor C need to be optimized.

This paper employs the SVM model for predicting the low-frequency component, and evaluates its strengths and weaknesses in conjunction with the physical interpretation of the low-frequency component. The ensuing analysis provides an assessment of the strengths and limitations of the SVM model:

Strengths:

- (i) Adaptation to long-term trends and periodic changes: Low-frequency components reflect long-term trends and periodic variations in electric, cooling and heating loads, and the SVM model is adept at handling such data. Through the learning of support vectors and the construction of a separating hyperplane, the SVM model effectively captures the long-term trends within low-frequency components, thereby enhancing prediction accuracy.
- (ii) Nonlinear modeling ability: The SVM model possesses robust capabilities for modeling nonlinear relationships, allowing it to effectively learn and represent complex nonlinear interactions within low-frequency components. This is particularly important for capturing nonlinear factors and interactions present in electric cooling and heating loads, enabling the SVM model to better fit and predict the changes observed in low-frequency components.
- (iii) Robustness and generalization capability: The SVM model exhibits robustness and strong generalization abilities when applied to electric cooling and heating load data. It effectively handles noise and outliers present in the data, and demonstrates the capacity to generalize predictions to unknown data. This characteristic plays a crucial role in ensuring the stability and reliability of low-frequency component predictions.

Limitations: Dependence on data standardization: The SVM model places high emphasis on data standardization. Given that low-frequency components may exhibit varying magnitudes and ranges of variation, data normalization becomes imperative to ensure the stability and performance of the model.

In conclusion, the SVM model possesses several advantages for predicting the low-frequency components of electric cooling and heating loads in integrated energy systems. It effectively adapts to long-term trends and nonlinear relationships, while exhibiting robustness and strong generalization abilities. However, it is crucial to address challenges such as parameter selection, high computational complexity, and data normalization. In practical applications, careful consideration of problem requirements and data characteristics is necessary, along with reasonable adjustment of model parameters and appropriate data processing, to achieve more accurate prediction results.

To demonstrate the suitability of SVM for handling low-frequency components, a low-frequency curve is selected for comparative analysis. Appendix Fig. A3 displays the fitting outcomes for each model. It is evident from the comparison of low-frequency components that SVM exhibits the highest level of fitting accuracy. Fig. 8 illustrates the error analysis, indicating that SVM has the lowest MAPE and RMSE values of 0.74% and 13.46, respectively. SVM possesses notable strengths in terms of robustness and its ability to control overfitting, along with its superior capability to extract and process low-frequency component features. The comparative analysis unequivocally reveals the inferior performance of other predictive models when confronted with low-frequency

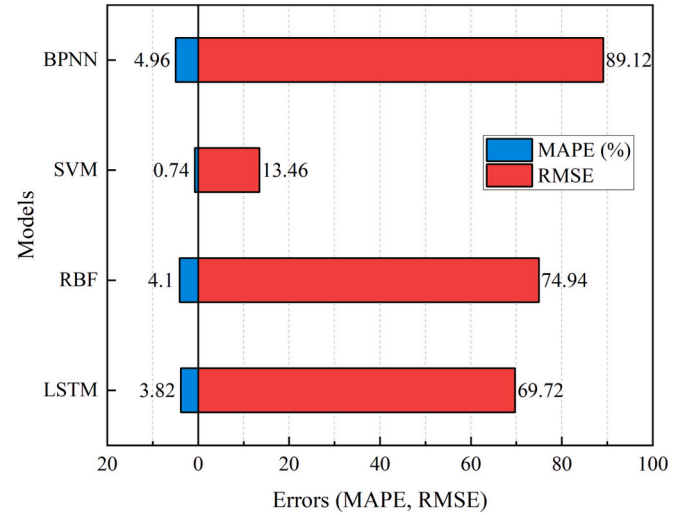


Fig. 8. Error analysis of four models at low-frequency.

components. In contrast, SVM stands out as an easily implementable approach that yields excellent prediction results when handling low-frequency components. Hence, this study opts for SVM as the preferred method for handling low-frequency components.

2.7.2. PSO-SVM prediction model

The particle swarm optimization algorithm is an optimization technique in which a group of particles randomly searches for the optimal value of the current optimal particle in a D-dimensional solution space. Through continuous iteration, the algorithm is able to converge to the global optimal solution [35]. During each iteration, each particle updates its velocity and position by tracking two extreme values: the first is the best solution found by the particle itself, and the second is the global best solution. The velocity and position of each particle are updated according to the following formula:

$$v_{id} = \omega v_{id} + c_1 r_1 (p_{id} - x_{id}) + c_2 r_2 (p_{gd} - x_{id}) \quad (32)$$

$$x_{id} = x_{id} + v_{id} \quad (33)$$

In (32) and (33): ω is an inertial weight factor that prevents local optimum and is usually an unfixed constant. c_1 and c_2 are the acceleration constants. r_1 and r_2 are random numbers between 0 and 1. x_{id} is the position of particle i in D-dimensional space. v_{id} is the velocity of particle i in D-dimensional space. p_{id} is the current optimal position of the population searched by the i -th particle. p_{gd} is the optimal position particle.

The update process of particles in the PSO algorithm involves randomizing the speed of each particle, which allows them to explore the search space. This feature enables the algorithm to achieve global optimization and maintain diversity during the optimization process. In the update formula (32), the term $(p_{id} - x_{id})$ represents the self-optimization process of the particle, as it calculates the distance between the current position and the particle's optimal position. On the other hand, the term $(p_{gd} - x_{id})$ calculates the distance between the current position and the optimal position of the group, which enables the sharing of information among particles.

The accuracy of prediction in SVM greatly depends on the selection of parameters C and σ . In order to optimize these parameters and obtain the best combination of them, particle swarm optimization is combined with SVM. The objective function of parameter optimization can be formulated as:

$$\min f(C, \sigma) = \sum_{i=1}^n (u_i - \hat{u}_i)^2, \text{ s.t. } \begin{cases} C \in [C_{\min}, C_{\max}] \\ \sigma \in [\sigma_{\min}, \sigma_{\max}] \end{cases} \quad (34)$$

In (34): u_i and $\hat{u}_i$ represent the output value of the i -th sample and the

predicted value of the model, respectively. The objective of SVM parameter optimization is to minimize the value of $f(C, \sigma)$ by identifying a set of parameters (C, σ) . The steps involved in the parameter optimization are as follows:

Step 1: Initialize the parameters, such as the size of the PSO population, the inertia factor, the number of iterations, and the acceleration factor. Set the initial values for the SVM parameters C and σ .

Step 2: Calculate the fitness value of each particle based on its current position, and track the best position found by each particle and the global best position found by the entire population.

Step 3: Update the velocity and position of each particle according to the formula (32) to generate a new population $x(t)$. Calculate the fitness value of the new population and compare it with the historical fitness value. If the current position is better than the historical best positions, update the best position of the particle and the global best position of the population.

Step 4: If the stopping criterion is met, terminate the optimization process and go to Step 5. Otherwise, go back to Step 2.

Step 5: Map the global optimal solution obtained from PSO to the SVM parameters to obtain the optimal model with the best combination of C and σ .

2.8. Considerations for distinct prediction methods

This paper employs distinct prediction techniques for high-, medium- and low-frequency, taking into account the following factors:

- (i) Different time scales consideration: The high-, medium-, and low-frequency components represent variations in the load data across different time scales. High-frequency components capture short-term fluctuations and rapid changes, medium-frequency components correspond to medium-term periodic changes, while low-frequency components capture long-term trends and gradual changes. Since these components exhibit distinct temporal characteristics, employing different forecasting methods can effectively accommodate their specific patterns of change.
- (ii) Consideration of different influencing factors: The high-, medium-, and low-frequency components can be influenced by distinct factors. High-frequency components are typically influenced by instantaneous factors and noise, medium-frequency components are influenced by seasonal factors and cyclical changes, while low-frequency components are affected by long-term trends and structural factors. Employing different prediction methods can effectively capture the characteristics and changing patterns of these components in relation to their respective influencing factors.
- (iii) Consideration of the predictive model's fitness: Different forecasting methods exhibit varying adaptability and effectiveness in handling data with different time scales and changing patterns. In the case of studying the regional integrated energy system, the high-frequency components representing short-term fluctuations, as decomposed within this system, are better suited for utilization of time series-based models. This paper specifically employs the LSTM neural network, which aligns well with capturing and predicting the dynamics of these high-frequency components. The intermediate frequency components, characterized by periodic changes and their association with electric, cooling and heating, require the input of high-dimensional feature vectors. These components are better suited for seasonal models. In this paper, the RBF model is chosen as it excels in handling high-dimensional feature vectors. The RBF model leverages its own function approximation capabilities to effectively capture nonlinear relationships within seasonal data. The low-frequency component typically represents the long-term trend and slow changes in the load data. In this paper, the SVM model is chosen to forecast this component. As a nonlinear classification and

regression method, SVM exhibits robust noise suppression capabilities and demonstrates good prediction performance. By selecting an appropriate forecasting method like SVM, the characteristics and parameters of the model can be effectively leveraged to enhance the accuracy of load forecasting.

- (iv) Distinguishing data characteristics: The high-, medium-, and low-frequency components exhibit distinct characteristics in terms of frequency domain, amplitude, fluctuation ranges, and variation ranges. By employing different prediction methods, these unique features can be effectively captured, enabling more accurate predictions of the behavior and patterns exhibited by each component.

2.9. Applicability comparison of optimization algorithms

The primary objective of optimization algorithms is to identify the best possible solution to a problem, either by maximizing or minimizing a specific objective function. Utilizing optimization algorithms allows for effective resource optimization, increased efficiency, cost reduction, improved decision-making, and various other advantages. Furthermore, these algorithms play a significant role in uncovering hidden patterns and regularities within vast datasets, thus supporting decision-making and prediction tasks. Nevertheless, various optimization algorithms possess their own unique strengths, and a thoughtful selection of the appropriate algorithm can significantly enhance the overall predictive performance of a model. Depending on the nature and requirements of the specific problem at hand, opting for a suitable optimization algorithm can enhance solution efficiency, yield superior outcomes, and fully leverage the algorithm's inherent advantages.

To demonstrate the suitability of the chosen optimization algorithm in this paper, it is more beneficial to employ GA and PSO for optimizing predictions involving high or low frequencies. Afterwards, proceed to randomly choose two sets of data, one containing high-frequency data consisting of 4000 points, and another containing low-frequency data also consisting of 4000 points. Split both datasets into training and test sets using an 8:2 ratio. Apply three optimization algorithms, namely GA, PSO, and Sparrow Search Algorithm (SSA). Compare the performance of GA-LSTM, PSO-LSTM, and SSA-LSTM on the high-frequency dataset. Similarly, evaluate the performance of GA-SVM, PSO-SVM, and SSA-SVM on the low-frequency dataset. The SSA algorithm, proposed in 2020 by Jiankai Xue et al. [36], draws inspiration from the foraging behavior of sparrows. By leveraging the self-organization and cooperation observed in sparrows during foraging, this algorithm strikes a harmonious balance between global exploration and local optimization.

This paper examines the algorithm's adaptability by assessing both the degree of fit and prediction time. In the high frequency component, the primary focus is on accuracy, while the low-frequency component is primarily evaluated based on prediction time. RMSE and MAPE are utilized to assess the fitting error. The fitting curve diagram and error analysis pertaining to the last 400 points in the test set are presented below, available for reference in Appendix Figs. A4–A6. Furthermore, Table 3 and Table 4 provide additional details on the error analysis.

The comparison of the three models in the high-frequency segment is depicted in Appendix Figs. A4 and A5. Additionally, the error analysis for this comparison can be found in Table 3. The presence of numerous nonlinear and intricate noises is frequently observed in high-frequency data, coupled with a high sampling frequency and small intervals between data points, resulting in increased discontinuity. These rapid

Table 3
Error analysis and computation time at high-frequency.

Forecasting model	MAPE/%	RMSE	Computation time/h
GA-LSTM	1.66	60.81	6.5
PSO-LSTM	3.57	85.94	2.3
SSA-LSTM	2.70	74.04	4.7

Table 4

Error analysis and computation time at low-frequency.

Forecasting model	MAPE/%	RMSE	Computation time/h
GA-SVM	0.69	12.13	5.6
PSO-SVM	0.36	5.47	1.5
SSA-SVM	0.52	8.32	3.6

fluctuations and high volatility significantly amplify the challenge of making accurate predictions. Hence, when forecasting high-frequency components, it is crucial to prioritize the fitting degree. By doing so, the overall accuracy of the multivariate load forecasting model can be enhanced. The GA-LSTM model demonstrates the highest level of fit, as evidenced by the lowest MAPE and RMSE values of 1.66% and 60.81, respectively. GA possesses the capability to explore extensive solution spaces by employing random and evolutionary operations such as selection, crossover, and mutation [37]. This makes GA particularly effective when tackling intricate high-frequency data prediction problems. By exploring diverse model and parameter combinations, GA exhibits adaptability and global search capabilities, enabling it to simultaneously explore multiple solution spaces while maintaining awareness of the global optimal solution. Moreover, GA exhibits a notable degree of robustness when confronted with initial problem conditions and input data noise. It can effectively handle irrelevant information and disturbances to a certain extent. This capability enhances GA's ability to assist LSTM in addressing complex and multi-peak high-frequency data prediction challenges. Consequently, this paper utilizes the GA algorithm to optimize the parameters of LSTM.

The comparison of the three models in the low-frequency segment is presented in Appendix Fig. A6, while the error analysis can be found in Table 4. Low-frequency data typically exhibits a long-term trend, stability, and minimal volatility, making it comparatively easier to process and predict. Consequently, when dealing with low-frequency data, it becomes crucial to assess the optimal prediction timeframe. This evaluation helps minimize time costs and enhance overall efficiency. The three models have attained a high level of fit, with PSO-SVM exhibiting the shortest optimization prediction time of just 1.5 h while maintaining high prediction accuracy. Parallelism in the PSO algorithm allows multiple particles to simultaneously explore the solution space. Each particle utilizes its current position and velocity to search for solutions, enabling the algorithm to explore multiple candidate solutions concurrently [38]. This parallel approach accelerates the search process and enhances the overall efficiency of the algorithm. The PSO algorithm possesses the self-adaptation characteristic, allowing it to perform a search that adapts to the demands and objectives of the problem at hand. Each particle adjusts its search direction and speed by exchanging and updating information with itself and neighboring particles. This iterative process enables the algorithm to progressively approach the optimal solution. By leveraging this adaptability, PSO can effectively expedite the search for improved solutions, ultimately reducing the overall search time. In comparison to other optimization algorithms, PSO requires fewer parameter settings, with the primary parameters being the number of particles, maximum iterations, and velocity weight. This reduced parameter count leads to decreased complexity and tuning time. These advantages of PSO are particularly beneficial for SVM in making low-frequency predictions and minimizing time-related costs. Hence, this paper adopts PSO as the optimization technique for SVM in handling low-frequency components.

3. Numerical analysis

The multivariate load data used in this paper was collected from the Tempe campus of Arizona State University in the United States [39], while the meteorological data was obtained from the National Renewable Energy Laboratory [40]. The weather data takes into account four dimensions, namely temperature, humidity, wind speed, and solar

radiation. Fig. 9 displays the waveform of the electric, cooling and heating load. Based on Fig. 9, it is evident that the electric, cooling load curves exhibit similar patterns, demonstrating high values in summer and low values in winter. In contrast, the heat load follows an opposite trend, with low values in summer and high values in winter. These observations suggest that the cooling and heating loads show strong seasonal complementarity and are highly correlated, thus highlighting the effectiveness of using a multivariate load forecasting method.

Divide the period between January 1, 2022, and December 31, 2022, into training and test sets with an 8:2 ratio and make hourly predictions with a step size of 1 h.

3.1. Multiple load decomposition and group processing

In the study of the RIES, it is essential to gather additional data information, thereby requiring a more extensive decomposition of IMFs. Conducting a larger amount of decomposition offers several advantages, which are as follows:

- (i) Obtain more comprehensive signal details: Increasing the amount of decomposition enables the original load sequence to be broken down into a greater number of eigenmode functions, with each function representing signal components across various frequency ranges. A larger amount of decomposition facilitates a more precise capture of subtle changes, rapid fluctuations, and transient behavior present within the original sequence. The focus of this paper on studying the RIES proves advantageous in acquiring regional data information and developing load forecasting models. By leveraging a higher level of decomposition, the forecasting model becomes better equipped to learn and effectively capture the frequency characteristics of the load, thus improving its load forecasting capabilities.
- (ii) To provide more characteristic information: Each eigenmode function can be viewed as a distinctive feature extracted from the original load sequence. By conducting a higher level of decomposition, more IMFs can be obtained, consequently offering additional characteristic information for load forecasting. This expanded set of IMFs enriches the input feature space of the prediction model and enhances the model's capability to capture load variations. Thus, increasing the decomposition amount improves the model's ability to effectively model and predict load changes.

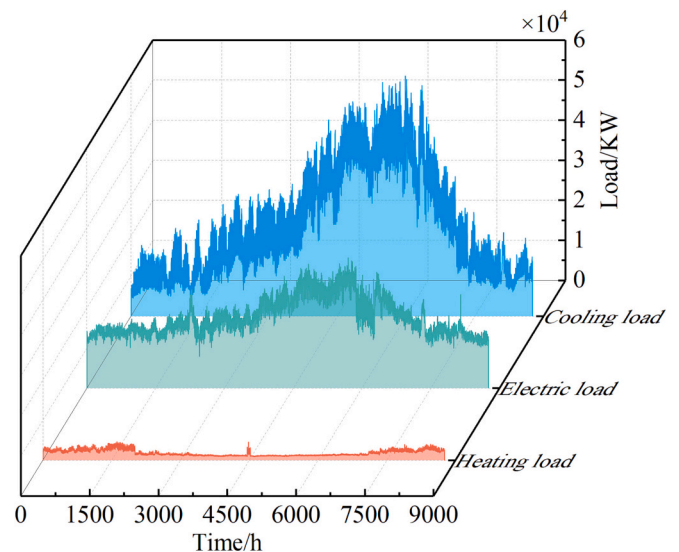


Fig. 9. Electric, cooling and heating load curve in 2022.

- (iii) Support for multi-frequency analysis: The research core of this paper revolves around the utilization of the CEEMD technique to decompose the original load sequence into a set of IMFs. By employing SE and zero-crossing rate, the IMFs are further classified into high-, medium-, and low-frequency components. Subsequently, suitable prediction models are selected to forecast each frequency component individually. This approach of conducting multi-frequency analysis and modeling plays a crucial role in enhancing the accuracy of load predictions.
- (iv) Mitigate the impact of noise and redundancy: Increasing the amount of decomposition allows for a more detailed signal decomposition, effectively dispersing noise and redundant information across different IMFs. This separation facilitates the identification and subsequent filtering of noise, leading to improved accuracy and stability in load forecasting.

Appendix Fig. A7 displays the decomposition results of the electric, cooling and heating load obtained by applying CEEMD to the initial load. CEEMD's high decomposition efficiency enables it to effectively reduce the non-stationarity of complex sequences and address the challenges in multivariate load forecasting for RIES. To achieve this, the key parameters of CEEMD, such as the Noise Standard Deviation ($Nstd$), Number of White Noise Groups (NR) and Maximum Iterations (Max_{iter}). Table 5 shows the specific parameters of CEEMD

After decomposing the electric, cooling and heating load into IMFs, the IMFs are grouped, and the zero-crossing rate is calculated for each IMF. In signal processing, it is often used to quantify the number of zero-crossing points in an audio or vibration signal. A zero crossing occurs when the signal changes polarity from positive to negative, or vice versa. By analyzing the zero-crossing rate of a signal, it is possible to distinguish between the high-, medium-, and low-frequency components. The zero-crossing rate of each component is defined as follows:

$$Z_j = \frac{j_{zero}}{J} \quad (35)$$

In (35): Z_j represents the zero-crossing rate, j_{zero} represents the number of zero-crossing points, and J represents the length of the signal.

By applying this formula, the zero-crossing rate of each IMFs for electric, cooling and heating can be calculated, as depicted in Fig. 10 and Table 6.

As per previous research [25], a zero-crossing rate below 0.01 is typically considered a low-frequency component. In this study, the IMFs with a zero-crossing rate below 0.01 were designated as low-frequency components, while those above 0.01 were classified as medium- and high-frequency components. Specifically, IMF9-IMF13 and RES were identified as low-frequency components, while IMF1-IMF8 were categorized as medium- and high-frequency components. In order to further separate the medium- and high-frequency, the sample entropy is analyzed next.

To group the medium- and high-frequency components more effectively, the sample entropy of all IMFs components generated by the decomposition was calculated and analyzed for stationarity. The sample entropy values of each IMF component are presented in Table 7. To provide a more intuitive description of the complexity of each component, a histogram of each sample entropy was plotted, as shown in Fig. 11. According to the literature [41], when grouping the IMFs based on sample entropy, a value >1.0 usually indicates a random component, which includes more high-frequency components. Combining this with

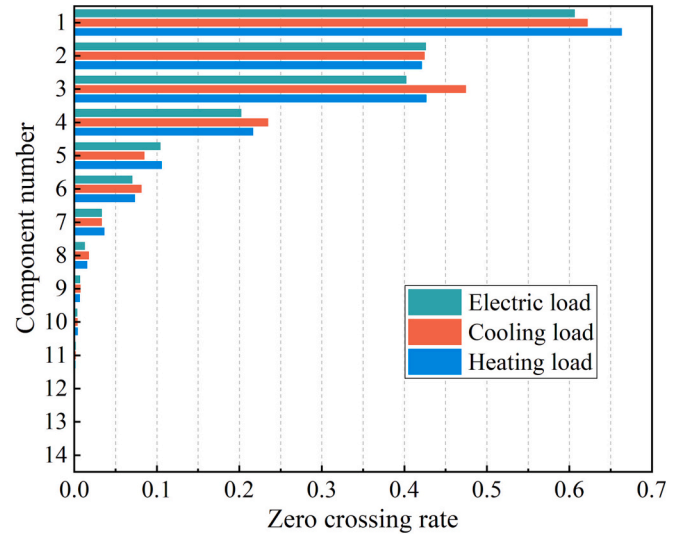


Fig. 10. Zero-crossing rate of electric, cooling and heating load IMFs.

the zero-crossing rate values of IMF1-IMF8 in Table 6, we observe that the zero-crossing rate and sample entropy of IMF5-IMF8 are both lower, while those of IMF1-IMF4 are higher. A higher sample entropy value indicates higher complexity, implying that the IMF component contains more nonlinear components, thereby making prediction more challenging. Therefore, IMF1-IMF4 are selected as high-frequency components, and IMF5-IMF8 are selected as medium-frequency components.

To group the electric cooling and heating load IMFs more effectively, the zero-crossing rate and sample entropy are combined, resulting in three groups of high-, medium-, and low-frequency components. Input vectors are constructed for the three sub-models based on this grouping.

3.2. Model parameter settings

The model receives 7 inputs which include electric, cooling and heating loads, temperature, humidity, wind speed, and solar radiation, and generates outputs for electric, cooling and heating loads. The proper parameter settings significantly affect the RIES multivariate load forecasting for each model, and Table 8 specifies the parameter settings for all the models.

3.3. Comparative analysis under different conditions

3.3.1. Modal decomposition comparative analysis

To assess the effect of Modal decomposition on the precision of RIES multivariate load forecasting, the forecast results for the period of November 4 to November 18, 2022 were randomly chosen for comparison. The contrast model excludes the CEEMD decomposition when selecting the predictive model for comparison. Analyze and contrast CEEMD-MF with LSTM, GA-LSTM, RBF-AE and PSO-SVM. The predicted results are presented in Appendix Fig. A8 and Fig. 12, with Appendix Fig. A8 providing an overall comparative analysis of 15 days, depicting the prediction outcomes of all five models. Fig. 12 shows two load series subsamples spanning one day, which are extracted from test set.

Fig. 12 consists of two groups of subplots: (a), (b) and (c) form the first group, while (d), (e) and (f) form the second group. These subplots demonstrate the fitting effect by comparing the prediction curves of different models with the actual loads. The curves are drawn in distinct colors. Fig. 12 (a)/ 12 (b)/ 12 (c) exhibit the comparison results for November 5, allowing for a visual comparison of the electric load and heating load among the prediction curves of various models. It is evident that the LSTM prediction curves, without CEEMD decomposition and hyperparameter optimization, has the poorest fitting degree with the

Table 5
CEEMD parameters.

Parameter name	Determined value
$Nstd$	0.5
NR	200
Max_{iter}	100

Table 6

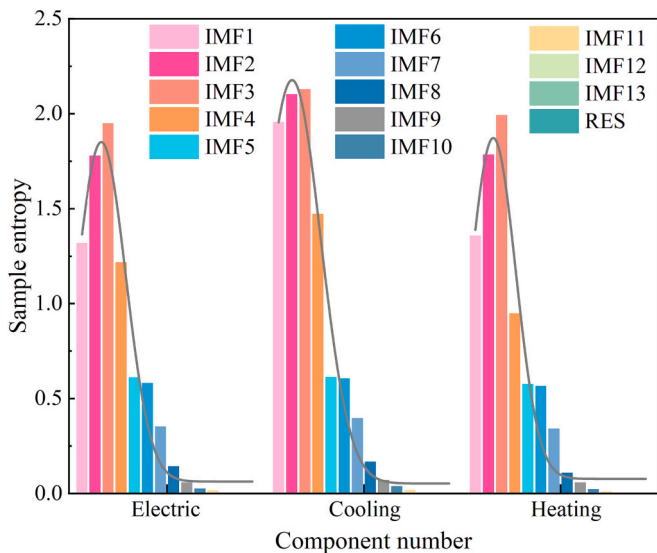
Zero-crossing rate of electric, cooling and heating IMFs.

Component	IMF1	IMF2	IMF3	IMF4	IMF5	IMF6	IMF7	IMF8	IMF9	IMF10	IMF11	IMF12	IMF13	RES
Electric	0.607	0.426	0.403	0.203	0.105	0.071	0.333	0.013	0.007	0.004	0.002	0.004	0.0009	0
Cooling	0.622	0.425	0.475	0.235	0.085	0.082	0.034	0.018	0.008	0.004	0.002	0.0007	0.0002	0
Heating	0.664	0.422	0.427	0.217	0.106	0.073	0.037	0.016	0.007	0.005	0.002	0.0006	0.0002	0

Table 7

Sample entropy value of electric, cooling and heating load.

Component	IMF1	IMF2	IMF3	IMF4	IMF5	IMF6	IMF7	IMF8	IMF9	IMF10	IMF11	IMF12	IMF13	RES
Electric	1.320	1.780	1.949	1.219	0.612	0.582	0.353	0.144	0.059	0.026	0.017	0.004	0.0015	0.0009
Cooling	1.956	2.102	2.129	1.473	0.614	0.607	0.397	0.168	0.072	0.039	0.018	0.0054	0.0014	0.0007
Heating	1.359	1.786	1.993	1.048	0.577	0.567	0.342	0.110	0.060	0.023	0.010	0.0051	0.0013	0.0006

**Fig. 11.** Sample entropy value of electric, cooling and heating load.**Table 8**

Main parameters setting for all models.

Modules	Parameter	Determined value
GA	DNA size	2
	Chromosome length	8
	Total group number	10
	Cross rate	0.6
	Mutation rate	0.01
GA-LSTM	Learning rate	0.01
	The number of LSTM layers	1
	The number of fully connected layers	1
	The number of neurons in the LSTM layer	182
	The number of neurons in the fully connected layer	158
RBF-AE	Dropout	0.2
	Number of hidden layer nodes	15
PSO	Learning rate	0.01
	Total group number	20
	Inertia Factor	2
	C	[0.1100]
	σ	[0.1100]
PSO-SVM	kernel function	RBF
	Particle Velocity Range	[0,1.0]
	C	5.8613
	σ	2.6222

actual load curve, particularly in the electric load curve between 5:00–20:00. On the other hand, the three sub-prediction models underwent hyperparameter optimization, but the fitting degree of the prediction curve without CEEMD decomposition is not as satisfactory as that of CEEMD-MF and the actual load. Fig. 12 (d)/ 12 (e)/ 12 (f) display the comparison results for November 14. The fitting effect of each model is akin to that of Fig. 12 (a)/ 12(b)/ 12(c), and the best fitting degree between the actual load and prediction curve is observed for CEEMD-MF. Based on the error results analyzed in Table 9 and Table 10, it is evident that CEEMD-MF exhibits the highest prediction accuracy, with the respective prediction errors of MAPE for electric cooling and heating loads being only 1.49%, 2.67%, and 1.60%. Hence, it can be concluded that modal decomposition has a significant impact on the accuracy of load prediction. Among the three loads, the electric load exhibits the highest prediction performance, whereas the cooling load has a relatively larger forecast error. This could be attributed to the decreasing demand for cooling load during autumn, which makes it more challenging to predict accurately. Overall, RIES multivariate load data exhibits significant fluctuations, making direct prediction less effective and resulting in relatively higher errors. However, modal decomposition can significantly improve the prediction accuracy.

3.3.2. Comparative analysis of CEEMD effectiveness

To assess the effectiveness and advantages of the proposed CEEMD decomposition method, two additional decomposition methods, namely EEMD and VMD, are introduced for comparative analysis. The parameter configurations for EEMD and VMD can be found in Tables A1 and A2 in the Appendix, respectively. Among these parameters, NE represents the number of EEMD Modes, α represents the balance parameter for data fidelity constraint, τ represents the time step, R represents the number of decomposition layers, DC represents synthetic signal and DC is set to true if the first mode is placed at DC (0 frequency). $Init$ represents the signal initialization, and tol represents the convergence fault tolerance criterion. Generally, except for R , which is the number of decomposition modes, the other parameters have empirical values associated with them. To ensure a fair comparison, the models generated by the three decomposition methods are subjected to prediction using a multi-model fusion approach. The EEMD and VMD decompositions undergo frequency division processing and are subsequently input into the three models for prediction. The fitting degrees of EEMD-MF, VMD-MF, and CEEMD-MF are then compared individually. The predicted results are presented in Appendix Fig. A9 and Fig. 13. Appendix Fig. A9 represents a comparative analysis of 15 days. Fig. 13 shows two load series subsamples spanning one day, which are extracted from test set.

Fig. 13 consists of two groups of subplots: (a), (b) and (c) form the first group, while (d), (e) and (f) form the second group. From Fig. 13, it is evident that the fitting effects of EEMD-MF and VMD-MF in Fig. 13(a)/ (d) are unsatisfactory. However, CEEMD-MF exhibits a significantly higher degree of fit to the actual load. This observation is further

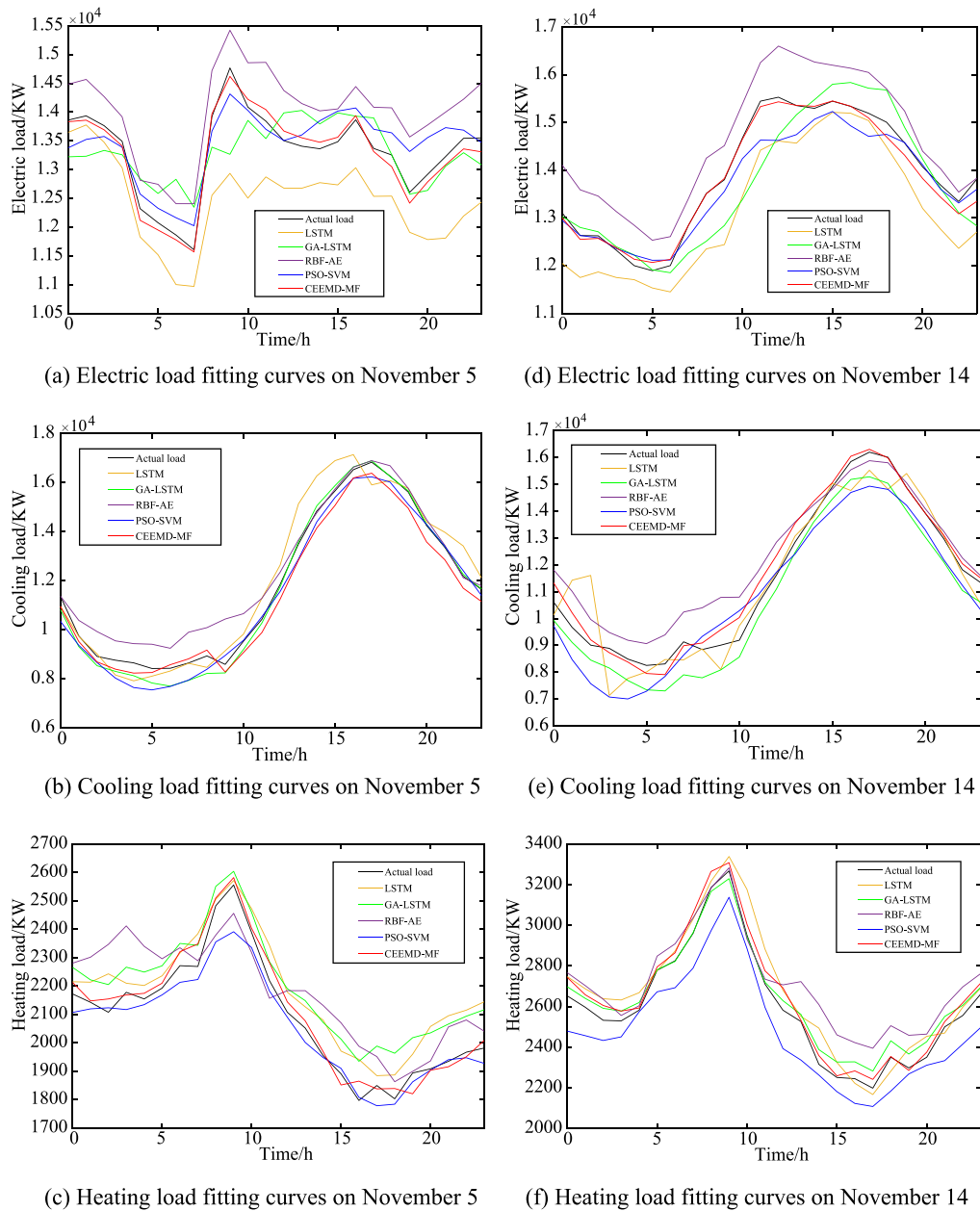


Fig. 12. The comparison between the predicted curve and the actual of models in two 24-h samples.

Table 9
MAPE comparison of forecasting models.

Forecasting model	Electric MAPE/%	Cooling MAPE/%	Heating MAPE/%
LSTM	5.61	5.75	4.40
GA-LSTM	3.60	5.16	3.40
RBF-AE	4.99	5.62	4.36
PSO-SVM	3.34	5.69	4.17
CEEMD-MF	1.49	2.67	1.60

supported by the error analysis presented in Table 11 and Table 12, where CEEMD-MF demonstrates the highest prediction accuracy. CEEMD, an improved version of EEMD, enhances the denoising ability and stability by introducing additional noise. Unlike EEMD, which introduces random noise components during iterative decompositions, CEEMD introduces more noise to achieve better denoising capabilities and stability. As a result, CEEMD provides clearer main modal components. Consequently, in the subsequent frequency division forecasting

Table 10
RMSE comparison of forecasting models.

Forecasting model	Electric RMSE/KW	Cooling RMSE/KW	Heating RMSE/KW
LSTM	785.57	1003.81	115.65
GA-LSTM	665.97	734.95	81.67
RBF-AE	495.74	760.81	119.14
PSO-SVM	867.70	875.02	130.30
CEEMD-MF	275.59	421.01	46.10

step, the combination model's advantages can be better utilized by the multivariate load forecasting method, leading to an overall improvement in forecasting accuracy.

VMD was initially designed to capture the primary frequency components of a signal using a reduced number of modal functions. However, a smaller number of modal functions implies a weaker ability to describe the frequency characteristics of the original signal.

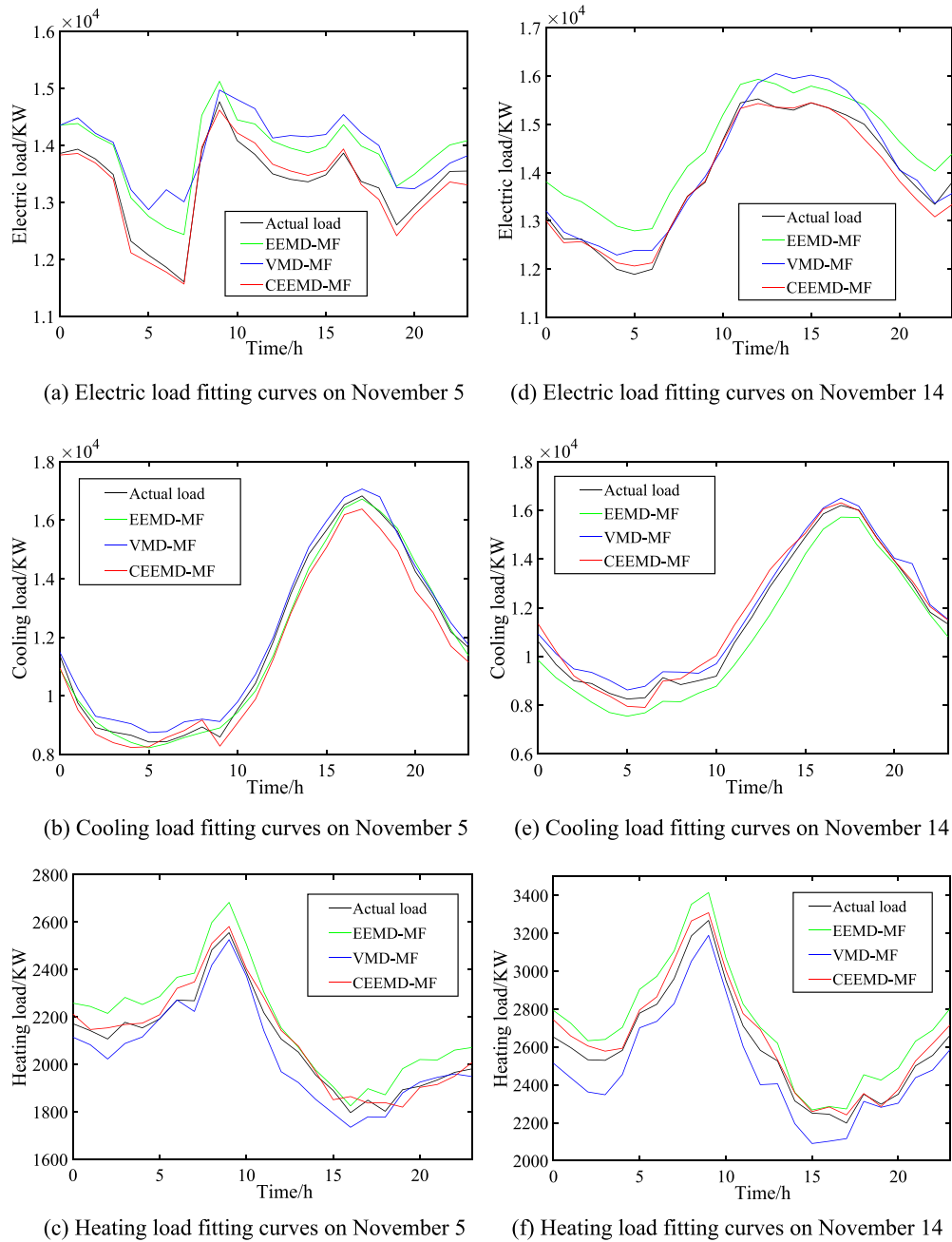


Fig. 13. The comparison between the predicted curve and the actual of models in two 24-h samples.

Table 11

MAPE comparison of forecasting models.

Forecasting model	Electric MAPE/%	Cooling MAPE/%	Heating MAPE/%
EEMD-MF	4.62	4.61	4.03
VMD-MF	4.13	3.22	3.54
CEEMD-MF	1.49	2.67	1.6

Table 12

RMSE comparison of forecasting models.

Forecasting model	Electric RMSE/ KW	Cooling RMSE/ KW	Heating RMSE/ KW
EEMD-MF	651.95	692.66	103.56
VMD-MF	697.37	445.37	96.28
CEEMD-MF	275.59	421.01	46.10

Consequently, in signals with rapid and significant fluctuations, certain details and frequency components are lost. This loss of information hampers the ability to achieve the desired prediction accuracy in subsequent frequency division forecasting. CEEMD generates multiple modal functions by utilizing multiple noise sequences, thereby producing a greater number of decomposition results to tackle subsequent prediction challenges. The decomposition results of VMD are influenced by parameters such as regularization parameters and bandwidth constraints. However, adjusting the regularization parameter can have different effects. Increasing the regularization parameter expands the smoothing limit on the data and decomposes more frequency components. However, a larger regularization parameter can also cause the modal functions to become excessively smooth, potentially leading to overfitting. In regards to CEEMD, which offers faster speed and lower computational complexity, this article devotes significant time to experimentation and fine-tuning during actual operations. This process

can be quite cumbersome, especially when handling RIES multivariate load data that demands more computing resources and time. Overall, this paper selects CEEMD as it provides a significant advantage in addressing the challenge of RIES multi-element load forecasting. The main factors behind this selection are the lower time cost and higher forecasting accuracy associated with CEEMD.

3.3.3. Multi-model fusion comparative analysis

To evaluate the impact of multi-model fusion on the accuracy of multivariate load forecasting, the comparison model selects the model that incorporates CEEMD decomposition and compares CEEMD-MF with CEEMD-LSSVM, CEEMD-GA-LSTM, and CEEMD-PSO-SVM. The predicted results are presented in Appendix Fig. A10 and Fig. 14. Appendix Fig. A10 represents a comparative analysis of 15 days. Fig. 14 shows two load series subsamples spanning one day, which are extracted from test set.

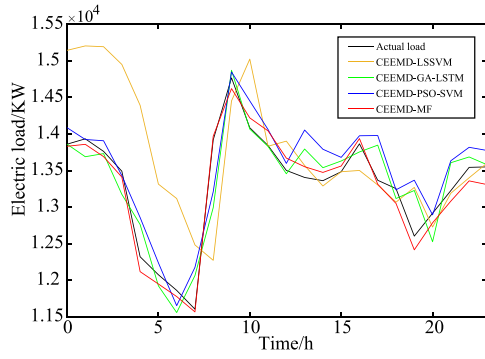
Fig. 14 consists of two groups of subplots: (a), (b) and (c) form the first group, while (d), (e) and (f) form the second group. The fit in Fig. 14

(a) is poor, especially for CEEMD-LSSVM in the time period 0:00–10:00. In contrast, only CEEMDSE-MF demonstrates the best fit with the actual load. The comparison results in Fig. 14(d)/(e)/(f) are consistent with that in Fig. 14(a)/(b)/(c), with CEEMD-MF displaying the best fitting degree with the actual load. Based on the analysis of errors presented in Table 13 and Table 14, CEEMD-MF has the highest prediction accuracy, while CEEMD-LSSVM has a larger prediction error. The use of a single prediction model in the three comparative models to predict the decomposed IMFs by CEEMD results in reduced prediction accuracy.

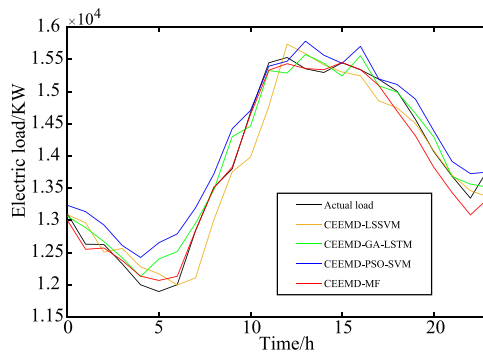
Table 13

MAPE comparison of forecasting models.

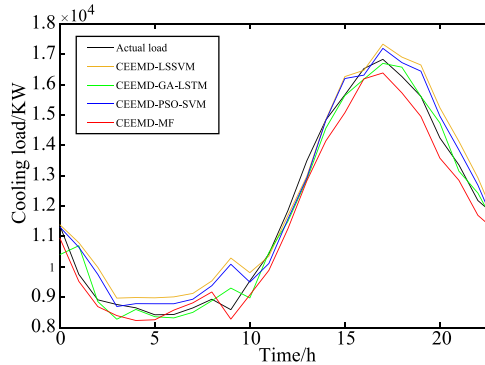
Forecasting model	Electric MAPE/%	Cooling MAPE/%	Heating MAPE/%
CEEMD-LSSVM	3.81	4.47	3.87
CEEMD-GA-LSTM	2.33	3.17	2.75
CEEMD-PSO-SVM	3.10	3.73	3.26
CEEMD-MF	1.49	2.67	1.60



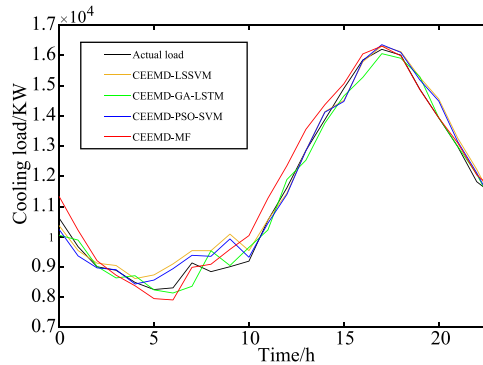
(a) Electric load fitting curves on November 5



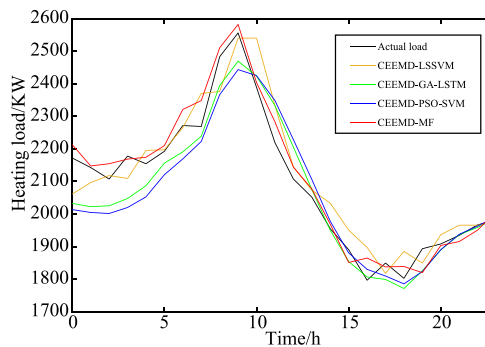
(d) Electric load fitting curves on November 14



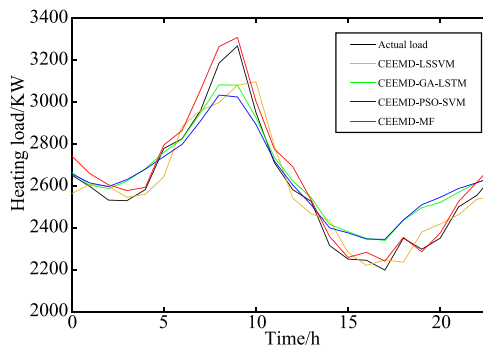
(b) Cooling load fitting curves on November 5



(e) Cooling load fitting curves on November 14



(c) Heating load fitting curves on November 5



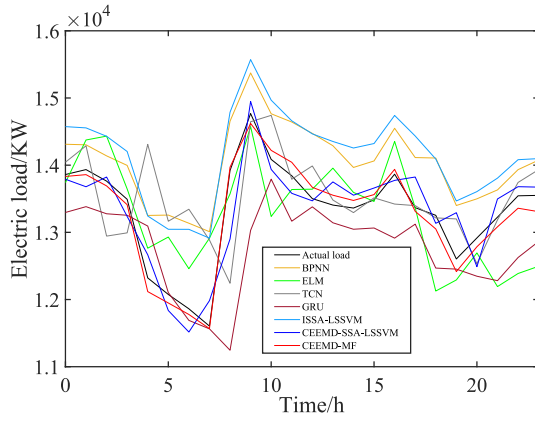
(f) Heating load fitting curves on November 14

Fig. 14. The comparison between the predicted curve and the actual of models in two 24-h samples.

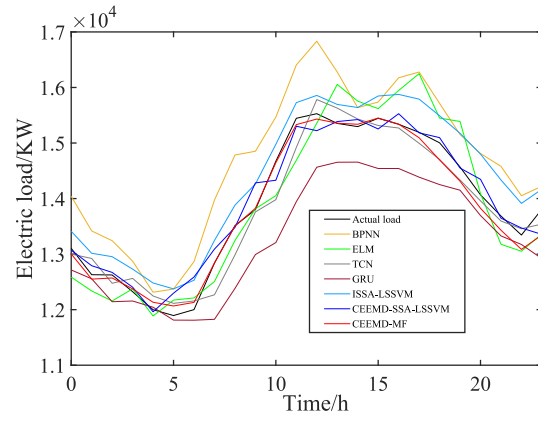
Table 14
RMSE comparison of forecasting models.

Forecasting model	Electric RMSE/ KW	Cooling RMSE/ KW	Heating RMSE/ KW
CEEMD-LSSVM	771.56	695.22	120.91
CEEMD-GA-LSTM	440.20	625.97	84.93
CEEMD-PSO-SVM	543.40	645.05	99.69
CEEMD-MF	275.59	421.01	46.10

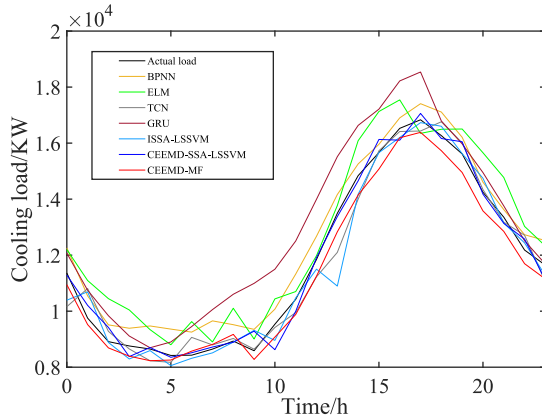
This is due to the fact that the decomposed IMFs components are not processed. Additionally, in the prediction of the RIES multi-element load, there are more IMFs components resulting from the decomposition of the three types of loads, thereby increasing the input dimension of the model. Therefore, further processing of the decomposed IMF is necessary. The model proposed in this paper employs an effective method to process the decomposed IMF. The process involves calculating the zero-crossing rate and sample entropy of the IMF, and dividing it into high-, medium-, and low-frequency components. These components are combined with the characteristics of the RIES data, and three models are used to handle each frequency component separately. The



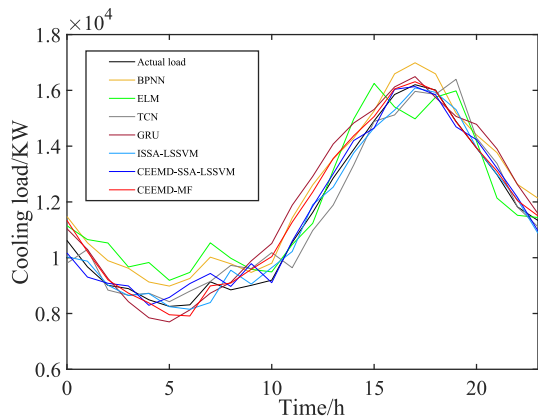
(a) Electric load fitting curves on November 5



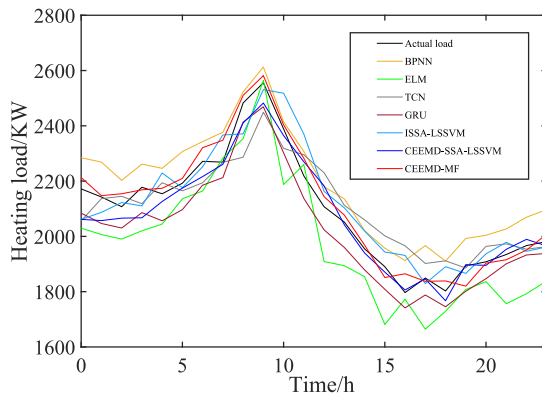
(d) Electric load fitting curves on November 14



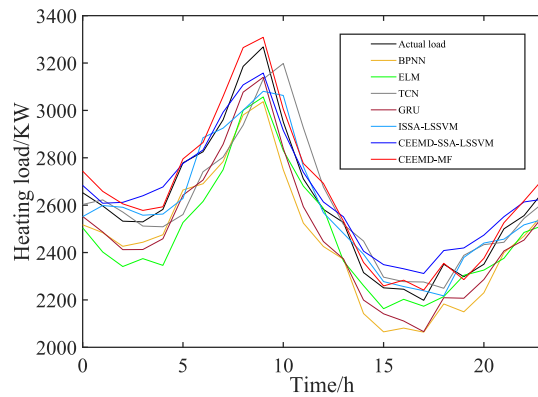
(b) Cooling load fitting curves on November 5



(e) Cooling load fitting curves on November 14



(c) Heating load fitting curves on November 5



(f) Heating load fitting curves on November 14

Fig. 15. The comparison between the predicted curve and the actual of models in two 24-h samples.

Table 15

MAPE comparison of forecasting models.

Forecasting model	Electric MAPE/%	Cooling MAPE/%	Heating MAPE/%
BPNN	3.85	5.94	3.62
ELM	3.66	6.27	6.05
TCN	3.53	4.98	4.18
GRU	4.90	5.64	4.54
ISSA-LSSVM	5.29	3.36	3.77
CEEMD-SSA-LSSVM	2.22	3.09	2.16
CEEMD-MF	1.49	2.67	1.60

Table 16

RMSE comparison of forecasting models.

Forecasting model	Electric RMSE/ KW	Cooling RMSE/ KW	Heating RMSE/ KW
BPNN	610.40	719.38	96.42
ELM	633.42	909.91	164.08
TCN	720.73	869.55	129.96
GRU	958.12	870.95	106.60
ISSA-LSSVM	753.15	656.56	117.09
CEEMD-SSA-LSSVM	413.81	500.46	65.70
CEEMD-MF	275.59	421.01	46.10

prediction results from these models are then reconstructed to obtain the final predictions. Based on comparative analysis, the proposed model demonstrates high prediction accuracy.

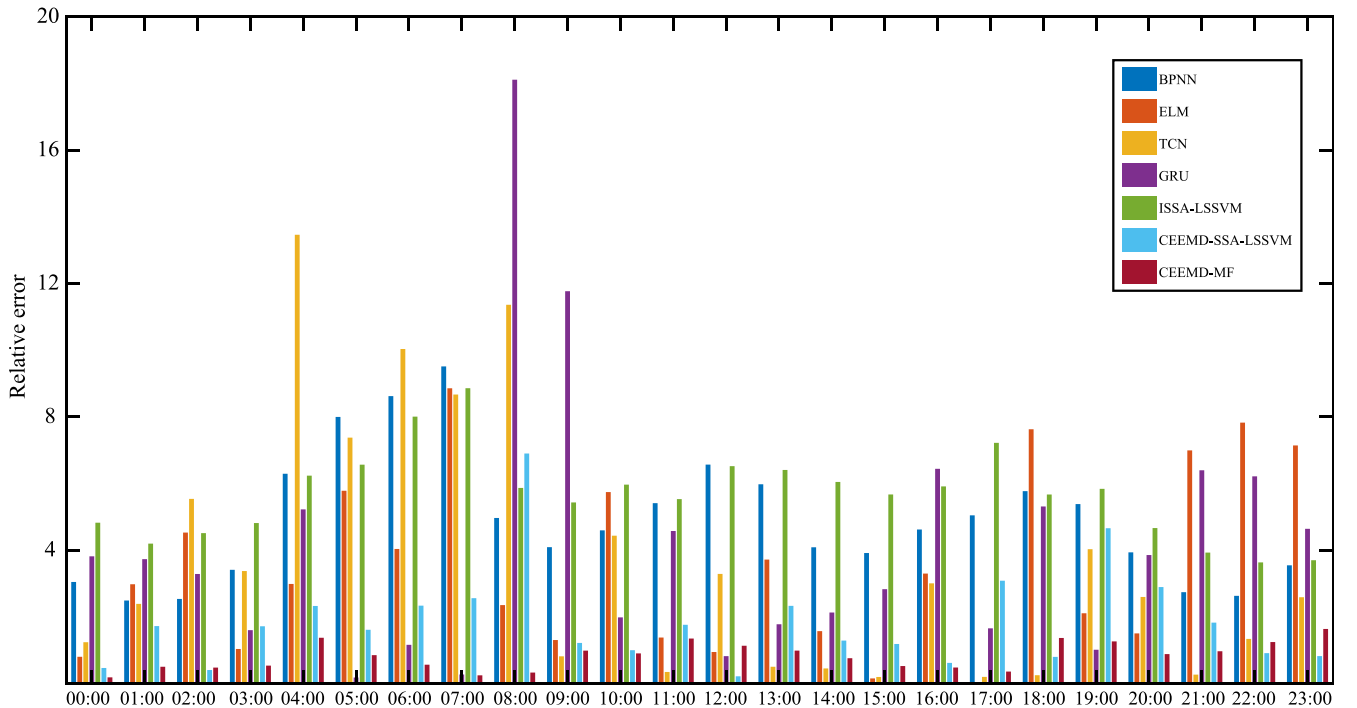
3.3.4. Comparative analysis of different forecasting models

The proposed model in this paper is compared with other forecasting models, such as neural networks, to validate its reliability and its ability to effectively enhance the accuracy of multivariate load forecasting. The models, namely BPNN, ELM, TCN, GRU, ISSA-LSSVM and CEEMD-SSA-LSSVM, are compared and analyzed against CEEMD-MF. Notably, ISSA-LSSVM employs an enhanced version of the sparrow algorithm to optimize its performance. The addition of Levy flight to the sparrow position update formula, introduced by ISSA, serves the purpose of reconnaissance and early warning. This enhancement significantly bolsters the

algorithm's optimization capability and its capacity to escape from local extreme values. Temporal Convolutional Network (TCN) is a specific type of neural network architecture designed for handling temporal data. On the other hand, GRU is a variant of recurrent neural networks and falls under the category of cyclic neural networks. Appendix Fig. A11 displays the predicted outcomes of the four models in 15 days comparative analysis, providing an overview of the fitting results. Fig. 15 shows two load series subsamples spanning one day, which are extracted from test set.

Fig. 15 consists of two groups of subplots: (a), (b) and (c) form the first group, while (d), (e) and (f) form the second group. To assess the reliability of the proposed model, commonly used models such as neural network BPNN, ELM, TCN, GRU, ISSA-LSSVM and CEEMD-SSA-LSSVM were selected for comparison. The proposed prediction model in this paper displays the best fit overall throughout the process. The results of Fig. 15(d)/(e)/(f) show a similar trend to Fig. 15(a)/(b)/(c), with CEEMD-MF maintaining a high degree of fitting. It is evident from Tables 15 and 16 that CEEMD-MF has the highest prediction accuracy, with the lowest MAPE and RMSE values. The BPNN, ELM, TCN, GRU, and ISSA-LSSVM prediction models, without undergoing modal decomposition, exhibit low prediction accuracy and poor fitting degree. The CEEMD-SSA-LSSVM model, as a standalone prediction model, lacks comprehensive processing of the decomposed IMF. It relies on direct superposition prediction, resulting in lower precision. Consequently, introducing multi-model fusion for prediction becomes more valuable in order to improve accuracy. Based on the findings presented in Table 15 and Table 16 of the error analysis, it can be observed that the GRU and ISSA-LSSVM models exhibit the highest forecasting errors in electric load forecasting. In the case of cooling load forecasting, the largest errors are observed with the BPNN and ELM models. Finally, for heat load forecasting, the ELM and GRU models demonstrate the highest forecast errors. CEEMD-MF achieves a lower MAPE of 0.73–3.8%, 0.42–3.6%, and 0.56–4.45% for electric, cooling, and heating loads, respectively. In summary, the CEEMD-MF prediction model described in the paper exhibits the highest level of prediction accuracy.

Ultimately, a relative error comparison is presented, focusing on the 24-h electrical cooling and heating loads of each model. (See Figs. 16–18)

**Fig. 16.** Comparison of electric load relative errors over a 24-h period.

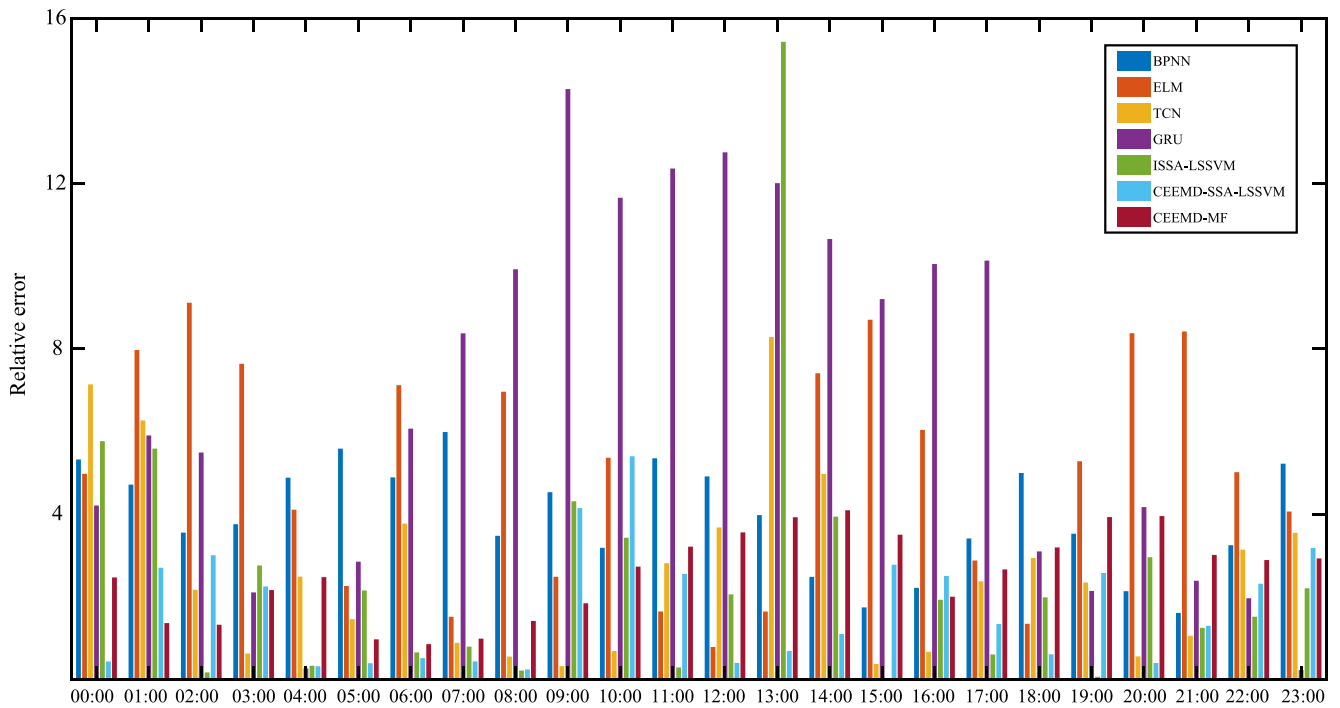


Fig. 17. Comparison of cooling load relative errors over a 24-h period.

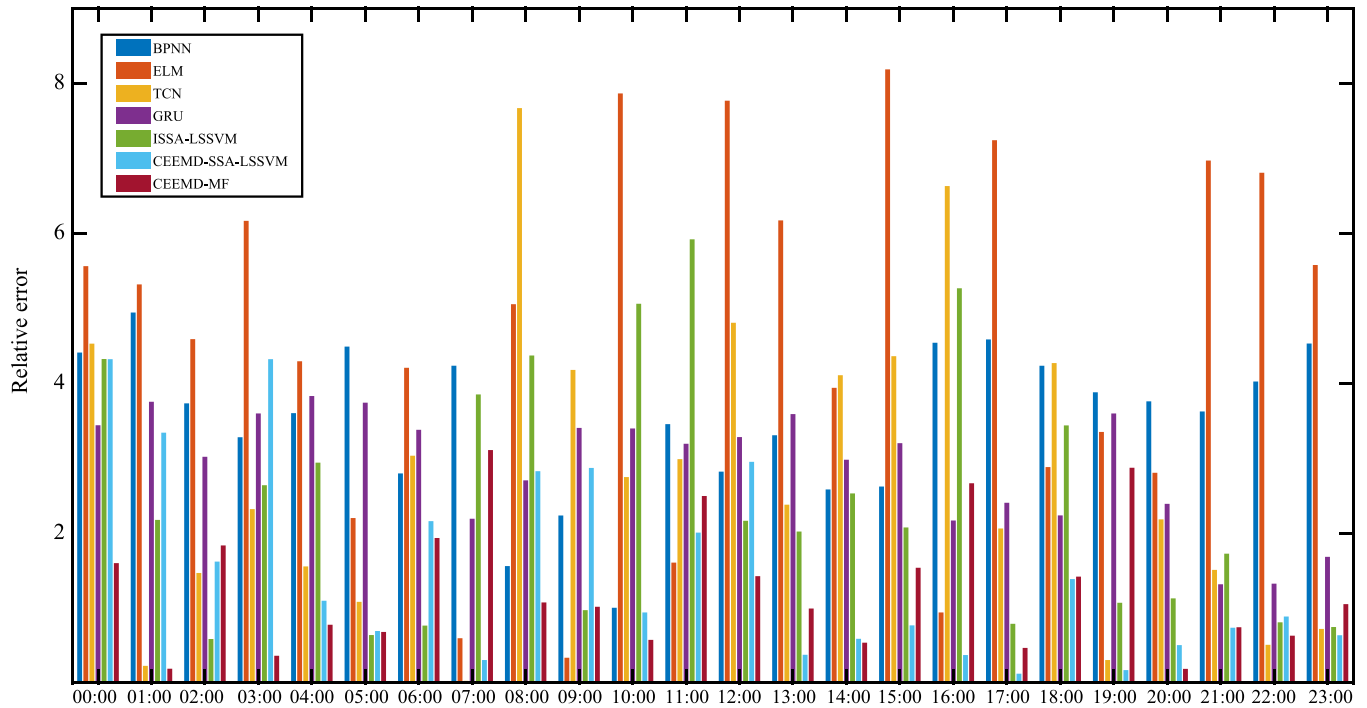


Fig. 18. Comparison of heating load relative errors over a 24-h period.

3.4. Discussions

The paper conducts simulation experiments from various perspectives, and the results confirm the effectiveness and high accuracy of the CEEMD-MF prediction model. Based on the information presented in the tables and figures, the following discussions can be drawn:

- (i) The conclusion can be drawn that the prediction accuracy of models that do not use CEEMD for load decomposition is

relatively poor. Based on the prediction results of different models, the CEEMD-MF proposed in this paper has the highest accuracy.

- (ii) Sub-prediction models demonstrated further improvement in prediction performance after undergoing hyperparameter optimization or model fusion dimensionality reduction processing. Specifically, these models displayed higher prediction accuracy in processing the corresponding frequency components.

- (iii) The multi-model fusion prediction method proves effective in handling non-stationary components obtained through CEEMD decomposition. The three sub-prediction models predict high-, medium-, and low frequency, which are then reconstructed to obtain the final prediction value. Comparative analysis results demonstrate that the CEEMD-MF model has the highest prediction accuracy and the smallest error.

Because RIES multivariate loads exhibit significant randomness and volatility, direct prediction leads to substantial errors. Thus, employing CEEMD to decompose the initial loads is essential for accurate forecasting. The decomposition results in a series of IMFs with different frequencies. In order to predict the IMFs with different frequencies effectively, it is necessary to select an appropriate prediction model for each component. To achieve this, the IMFs are categorized into high-, medium-, and low-frequency components by evaluating their zero-crossing rate and sample entropy. This division enables the identification of the most suitable prediction model for each specific frequency component. For the high-frequency components, LSTM is chosen as the preferred prediction model. LSTM units excel at capturing long-term dependencies in sequential data, making them well-suited for handling high-frequency data that exhibit time series characteristics. The medium-frequency component is predicted using the RBF, which is known for its strong performance in high-dimensional spaces. In RIES multivariate load forecasting, accurate predictions rely on using high-dimensional feature vectors as input for forecasting. This makes RBF neural networks an excellent choice for modeling and forecasting. SVM is selected for low-frequency components. SVM demonstrates good generalization ability and robustness in predicting low-frequency components. The RIES multivariate load sequence often contains outliers and noise, and SVM can effectively mitigate the impact of such noise and outliers on the prediction results. After undergoing hyperparameter optimization and model fusion, the three selected models were utilized for processing specific frequency components. Finally, the prediction results of these sub-models were reconstructed to obtain the final forecast value. Following a thorough comparative analysis, the multi-model fusion approach demonstrated higher prediction accuracy.

4. Conclusions

To address the challenge of forecasting RIES multivariate loads with high volatility and randomness, this study proposes a load forecasting approach that uses CEEMD and multi-model fusion. The conclusions are as follows:

- (i) In this paper, a novel data processing method is proposed, which combines the CEEMD frequency division technique with SE and zero-crossing rate. This method effectively addresses the challenge of predicting data with high volatility and instability, leading to a notable improvement in prediction accuracy. When compared to the prediction method that does not utilize modal decomposition, the CEEMD-MF approach demonstrates a reduction in MAPE for electric, cooling and heating by 1.85–4.12%, 2.49–3.08%, and 1.8–2.8%.
- (ii) To assess the efficacy and advantages of CEEMD, this study incorporates EEMD and VMD decompositions, comparing CEEMD-MF with EEMD-MF and VMD-MF. The results table shows that CEEMD-MF outperforms the other two decomposition prediction methods in terms of MAPE for electric load, cooling load, and heating load, exhibiting reductions of 2.64–3.13%, 0.55–1.94%, and 1.94–2.43% respectively.
- (iii) In this paper, a combined forecasting approach is employed, wherein different frequency components are selected and paired with suitable models. The resulting MAPE for electric, cooling and heating load is observed to be 0.84–2.32%, 0.5–1.8%, and 1.15–2.27% lower than the MAPE of the single load forecasting method following modal decomposition.
- (iv) The effectiveness of the proposed CEEMD-MF multivariate load forecasting method is demonstrated through a comparison with various other forecasting models including BPNN, ELM, TCN, GRU, ISSA-LSSVM, and CEEMD-SSA-LSSVM. The results indicate that the CEEMD-MF method consistently outperforms these models, resulting in a reduction in MAPE for electric cooling and heating by 0.73–3.8%, 0.42–3.6%, and 0.56–4.45%, respectively.

CRedit authorship contribution statement

Jian Shi: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.
Jiashen Teh: Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A

Refactoring pseudocode for predicted values at different frequency

Input: Predicted value of high-frequency component: $P_{IMF_{high}}$
 Predicted value of medium-frequency component: $P_{IMF_{medium}}$
 Predicted value of low-frequency component: $P_{IMF_{low}}$

Output: Load prediction P_{load} .

1: Initialize the load value $P_{load} = 0$.

2: Add the predicted values of the high-frequency, medium-frequency, and low-frequency IMF components together.

$$P_{load} = P_{IMF_{high}} + P_{IMF_{medium}} + P_{IMF_{low}}$$

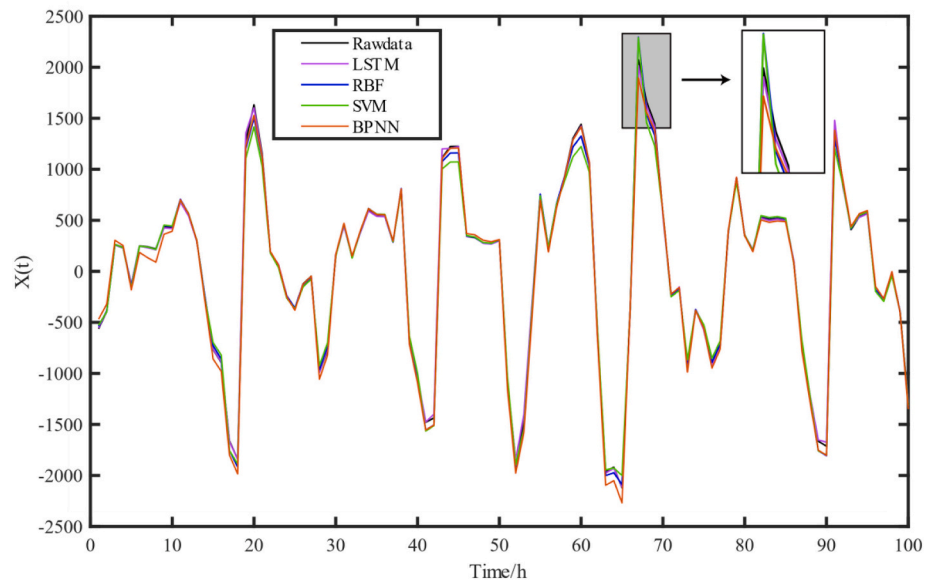


Fig. A1. High-frequency comparison of four models.

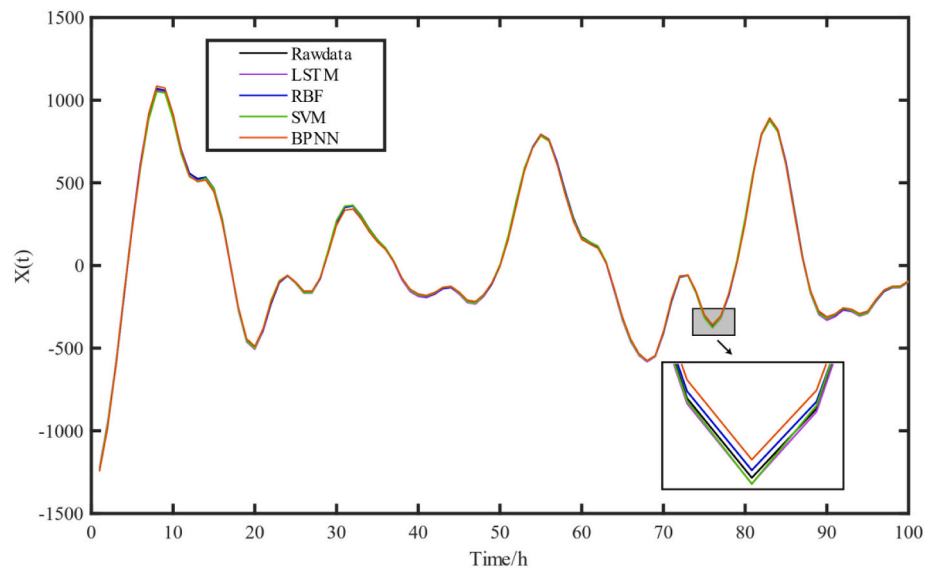


Fig. A2. Medium-frequency comparison of four models.

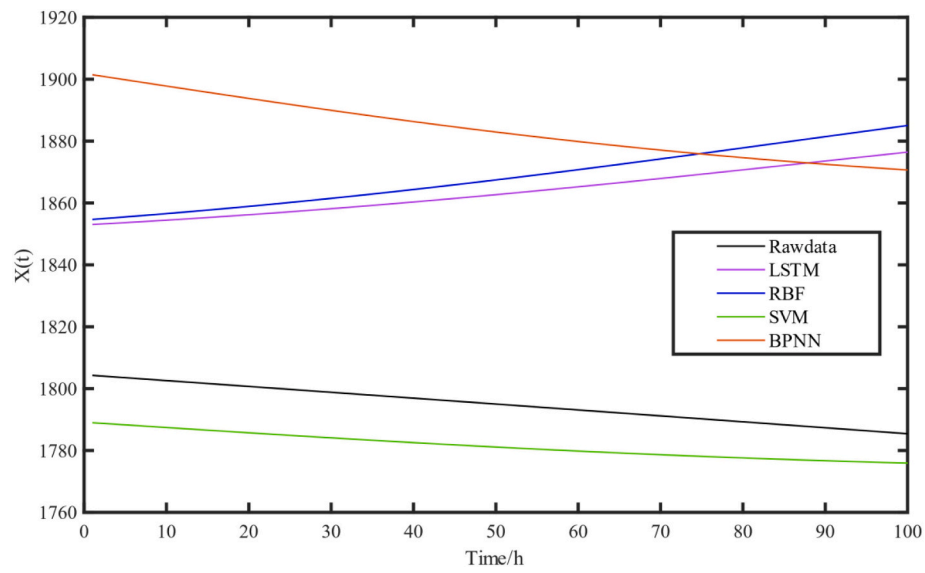


Fig. A3. Low-frequency comparison of four models.

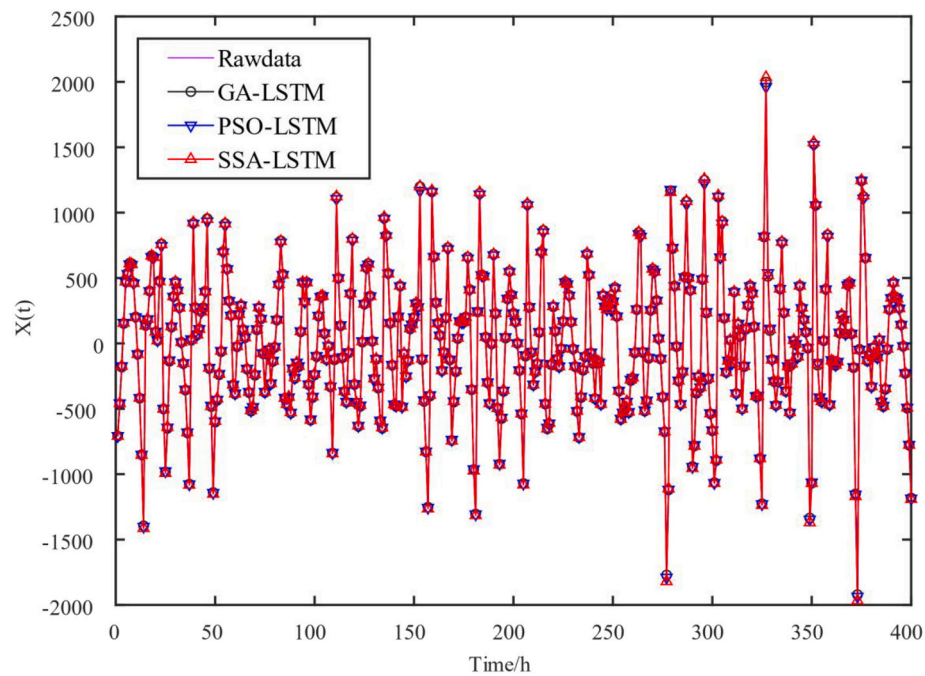


Fig. A4. Comparison of three optimization algorithms at high-frequency.

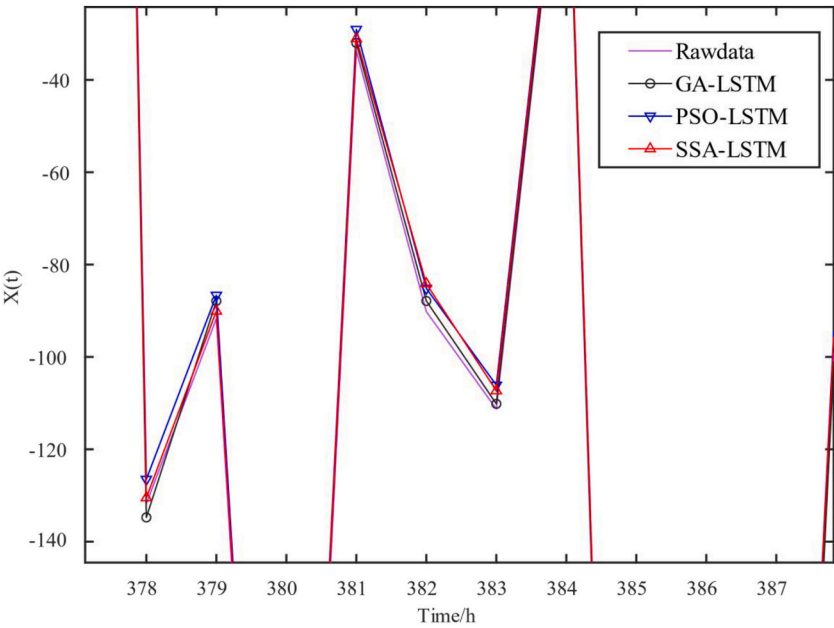


Fig. A5. Local comparison of three optimization algorithms at high-frequency.

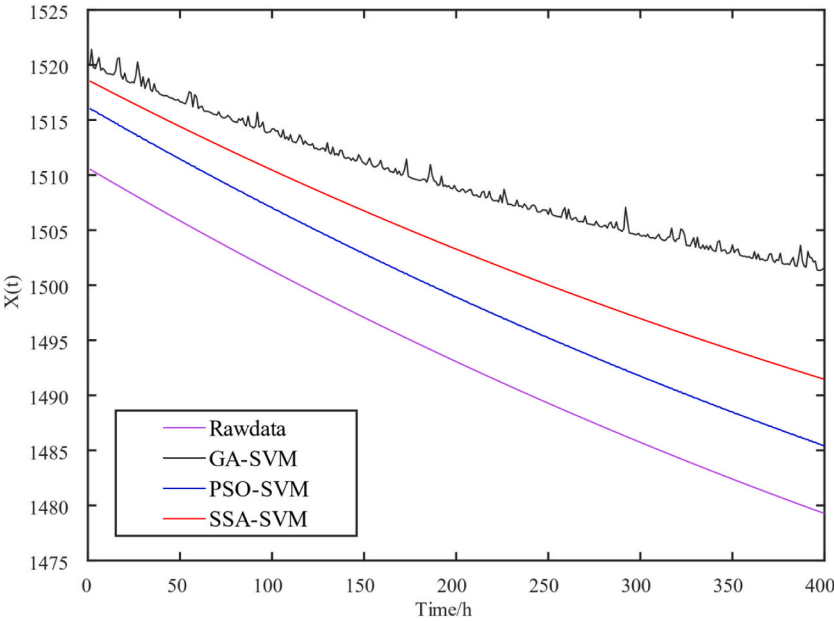
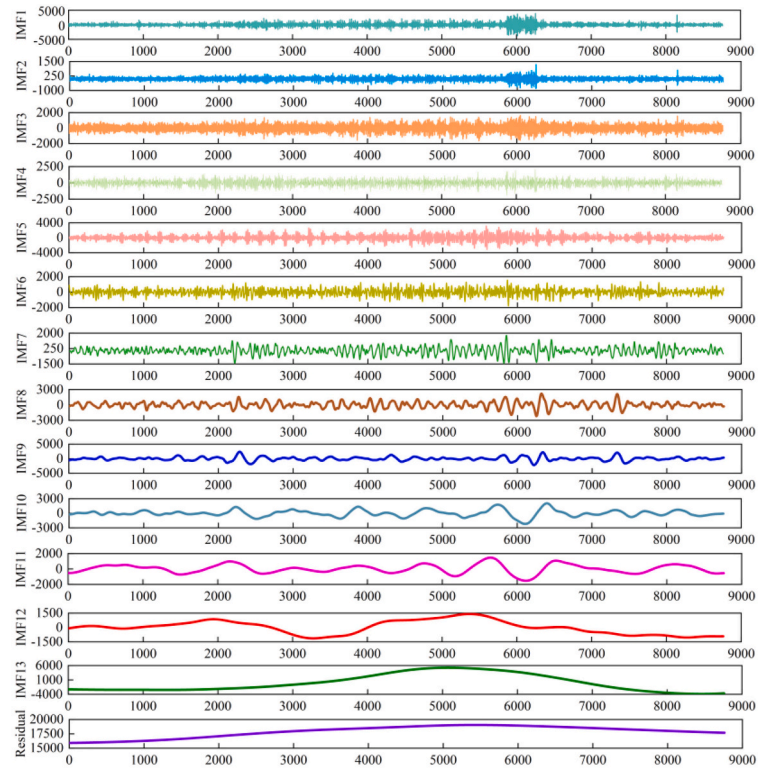
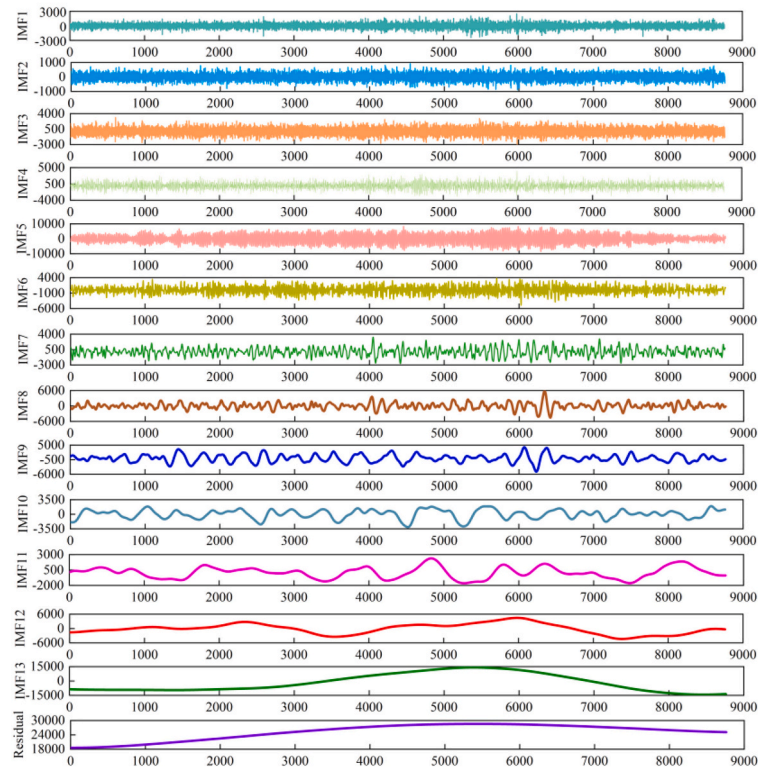


Fig. A6. Comparison of three optimization algorithms at low-frequency.

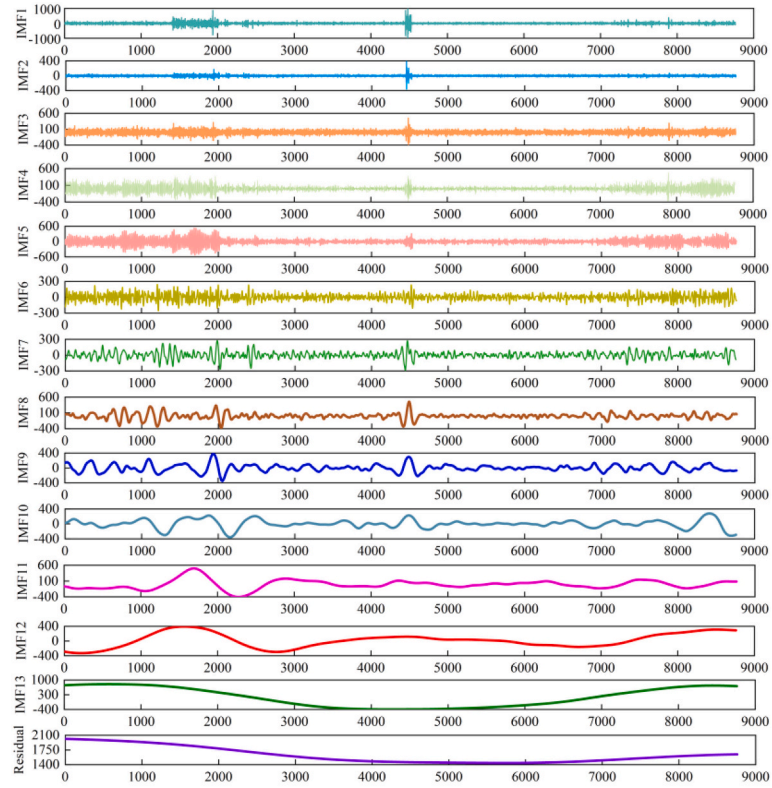


(a) Electric load component



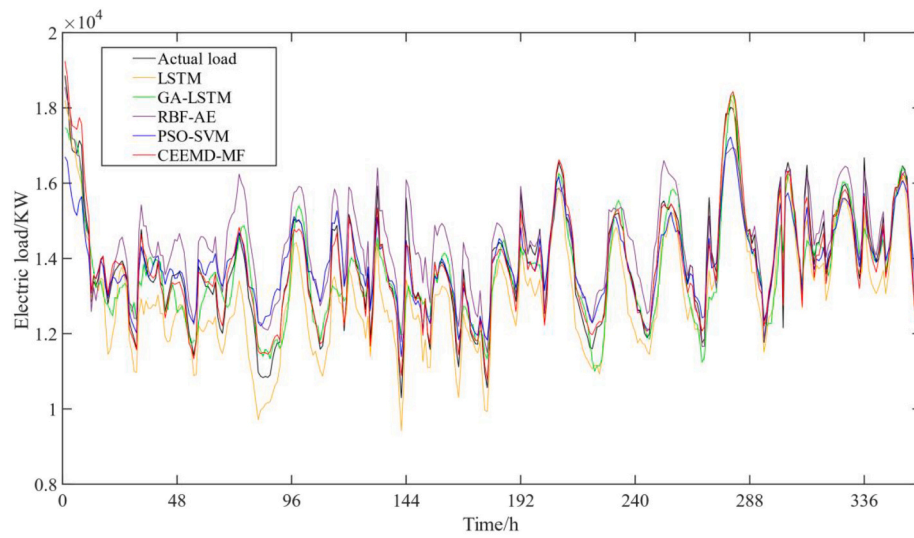
(b) Cooling load component

Fig. A7. (a) Electric, (b) Cooling and (c) Heating load component.

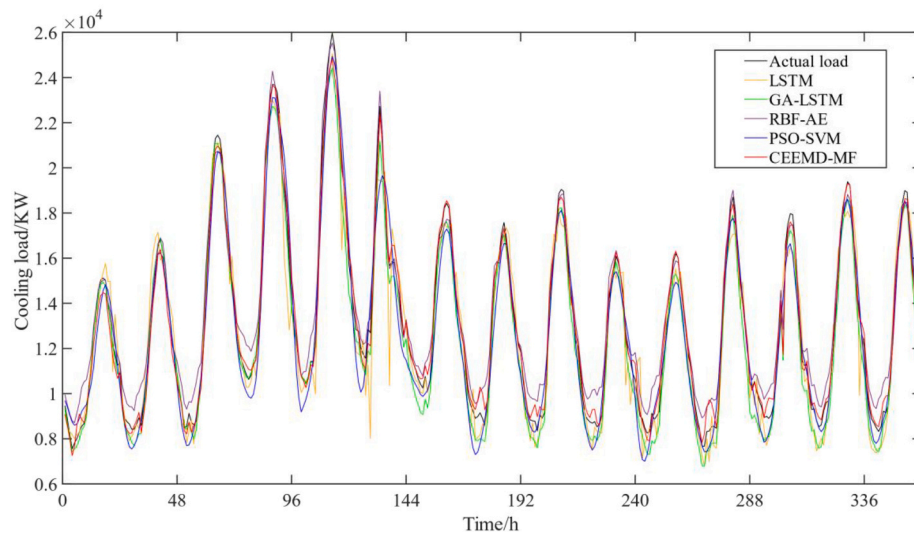


(c) Heating load component

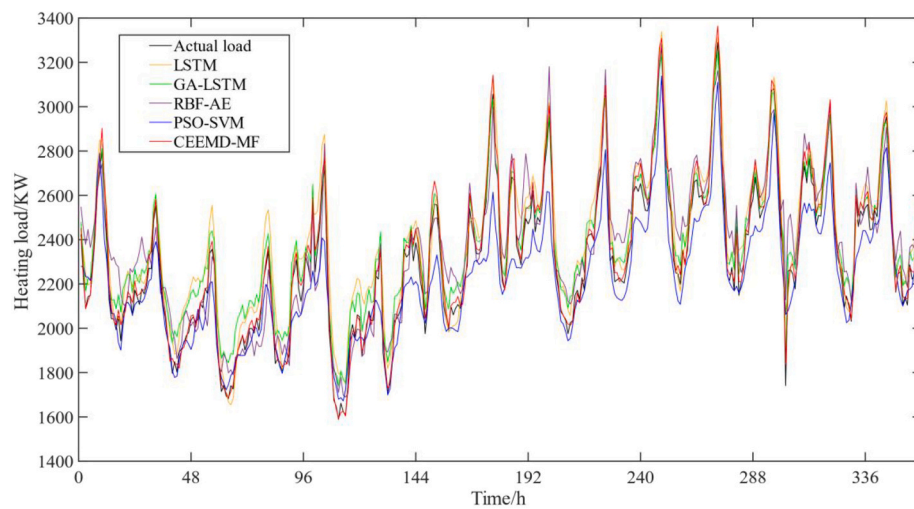
Fig. A7. (continued).



(a) Electric load

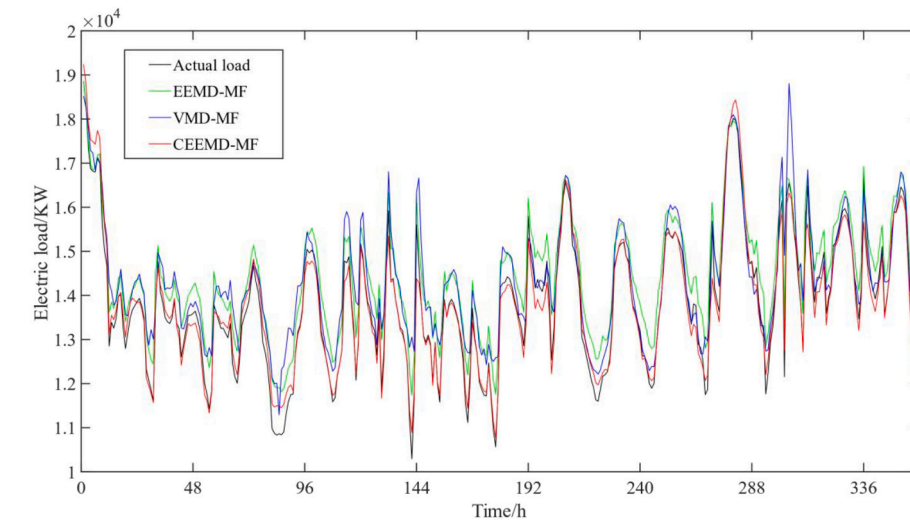


(b) Cooling load

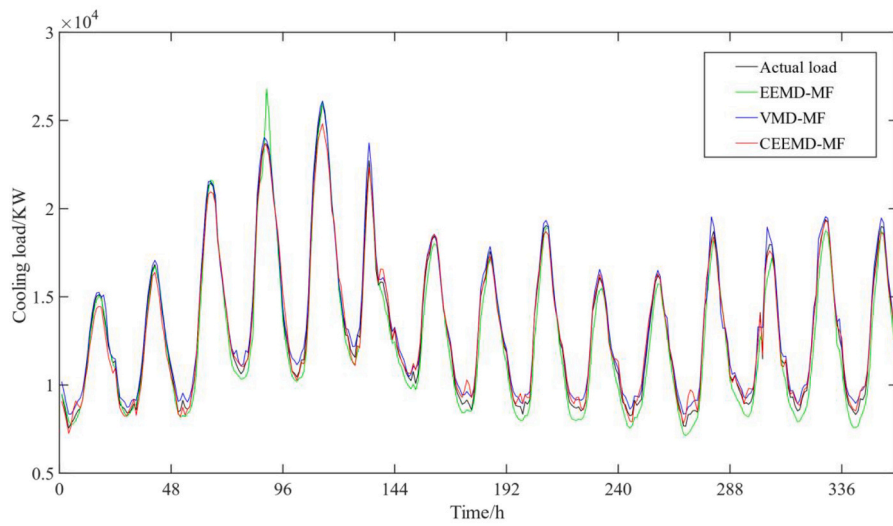


(c) Heating load

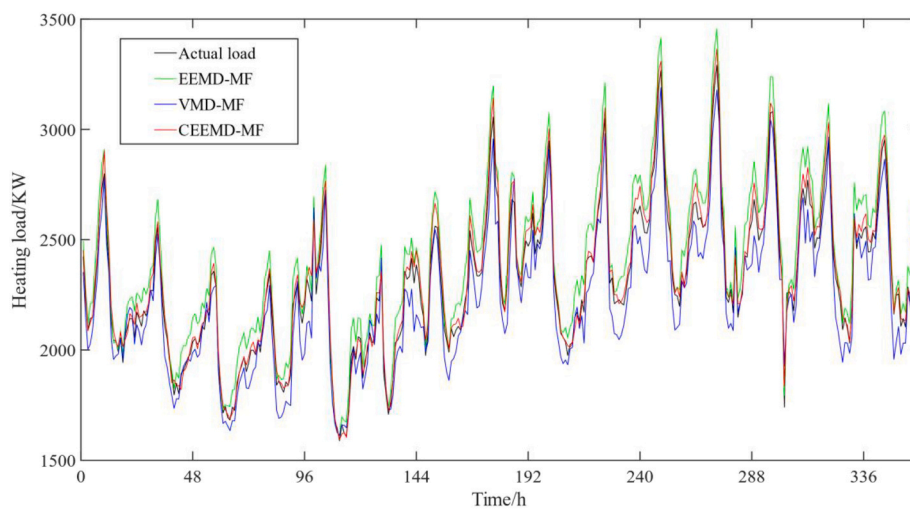
Fig. A8. The comparison between the predicted curve and the actual of models.



(a) Electric load

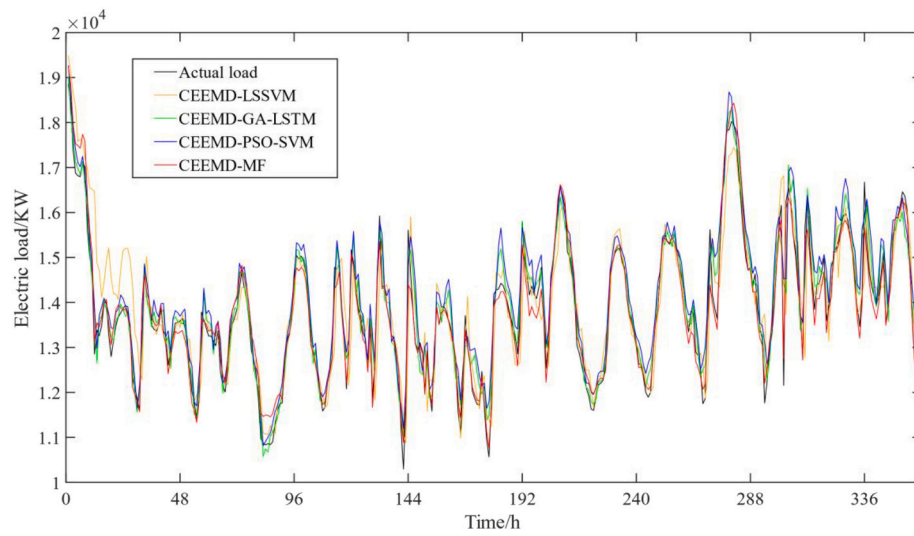


(b) Cooling load

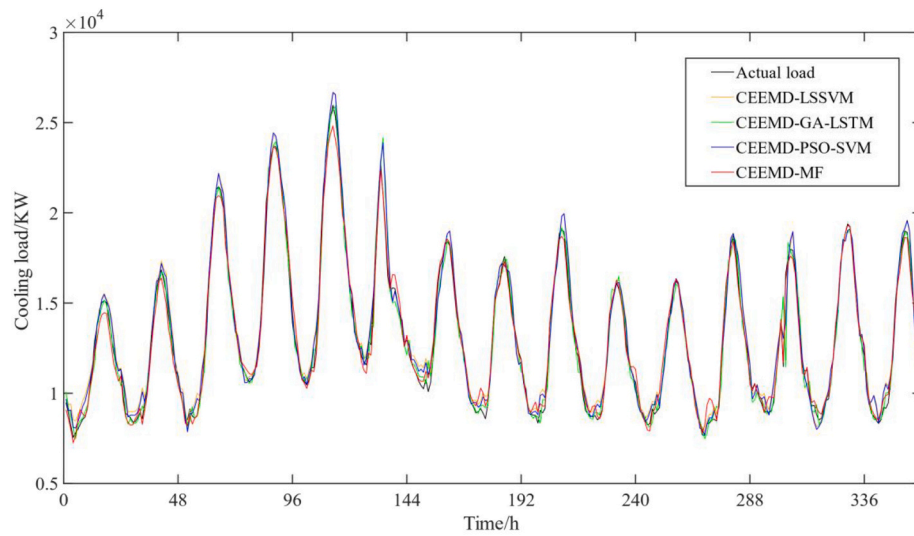


(c) Heating load

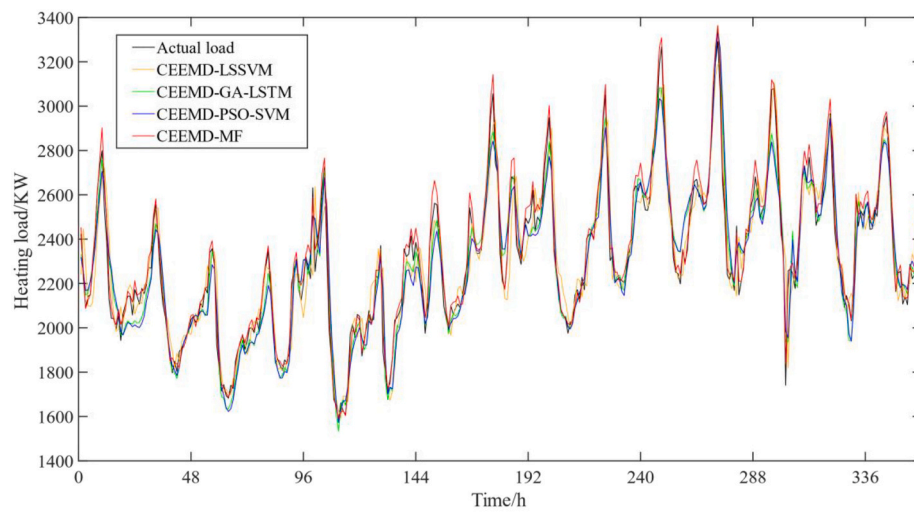
Fig. A9. The comparison between the predicted curve and the actual of models.



(a) Electric load

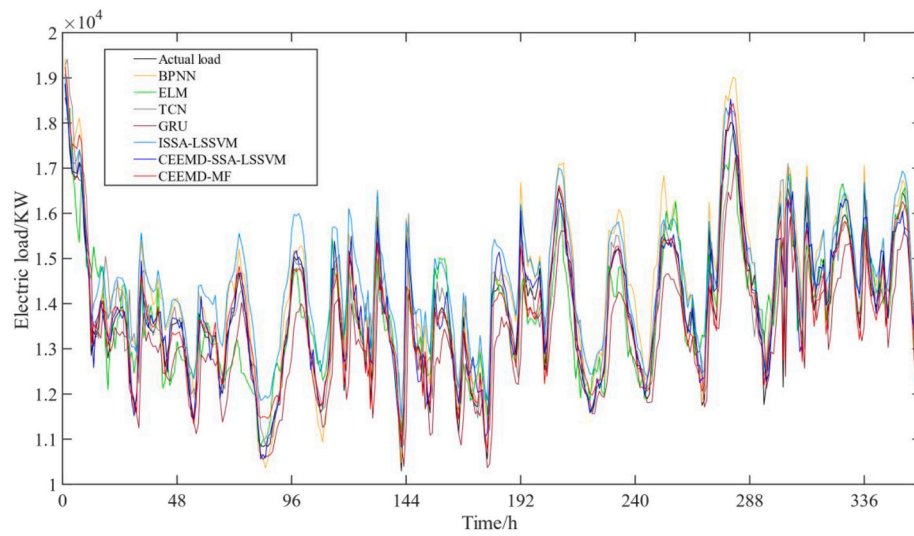


(b) Cooling load

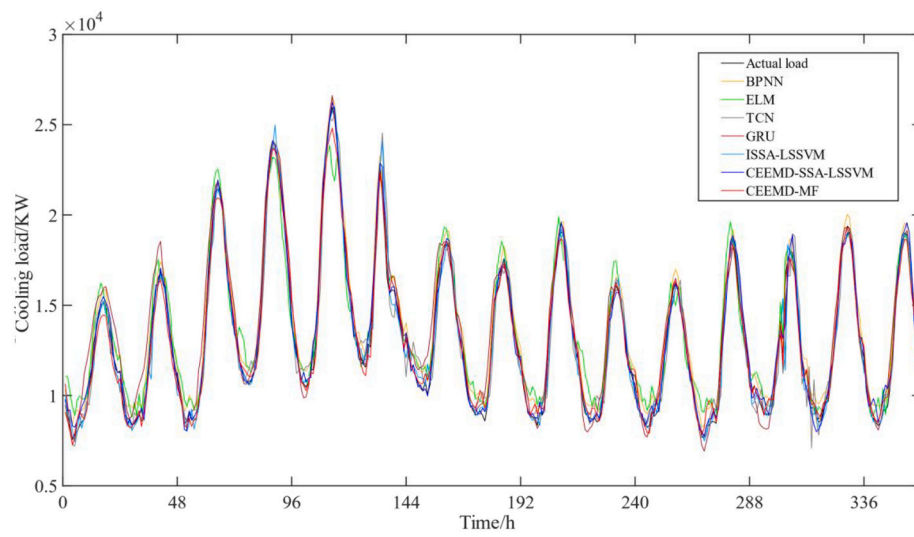


(c) Heating load

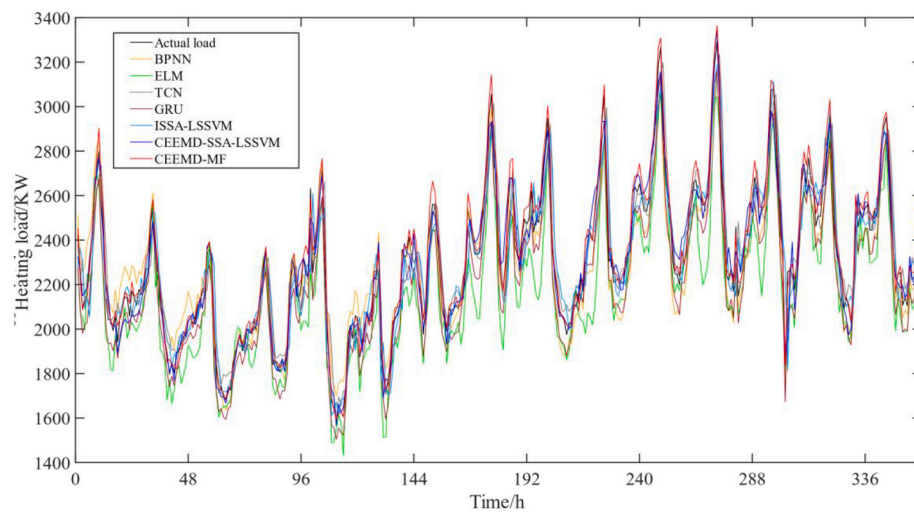
Fig. A10. The comparison between the predicted curve and the actual of models.



(a) Electric load



(b) Cooling load



(c) Heating load

Fig. A11. The comparison between the predicted curve and the actual of models.

Table A1
EEMD parameters.

Parameter name	Determined value
<i>Nstd</i>	0.1
<i>NE</i>	100

Table A2
VMD parameters.

Parameter name	Determined value
<i>alpha</i>	100
<i>tau</i>	0
<i>R</i>	5
<i>DC</i>	0
<i>init</i>	1
<i>tol</i>	1e-7

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